Data Sharing in Credit Markets with Collateral*

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Abstract

Data sharing–such as open banking initiatives–and improvements in data analytics enhance the capabilities of financial technology (fintech) companies and can reduce information asymmetries. This is generally believed to alleviate adverse selection and improve welfare. Traditional lenders such as banks, however, respond to the increasing competition by offering more attractive products that may involve costly collateral. We uncover a novel trade-off between improved information and the destructive competition induced by increased collateralization. We show that open banking and advances in data analytics may harm not only social welfare but also fintechs themselves, particularly when traditional lenders are sufficiently proficient at collateralization. We also examine alternative institutional arrangements, such as the allocation of property rights and the creation of data markets, that can outperform open banking. Our results contribute to the ongoing policy debate on the welfare implications of data-sharing initiatives.

KEYWORDS: Data sharing, open banking, fintechs, banks, collateral, welfare

JEL CLASSIFICATION: D82, L13, G21, G23

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1 Introduction

It is well established that financial markets are characterized by asymmetric information, which often leads to distortions and inefficiencies. In their seminal paper, Stiglitz and Weiss (1981) show that adverse selection can lead to credit rationing and potential underinvestment.[1] A substantial body of subsequent theoretical and empirical research has confirmed and extended these insights, highlighting the central role of information asymmetries in financial frictions. As a result, the production and dissemination of information that alleviates these asymmetries can have profound effects on market efficiency and investment outcomes.

Improvements in information and prediction arise from various sources. Perhaps the most important is the accumulation and dissemination of personal and other types of data, alongside the expansion of data centers, more powerful processors, and larger storage capacities. The countless online activities and digital connections that take place each day give rise to vast amounts of data that have reshaped the economy. The rise of artificial intelligence (AI) and data analytics has further accelerated this transformation. Better-trained algorithms allow lenders to make more accurate predictions about potential borrowers. In this new landscape, financial technology companies (fintechs) leverage technology and the increasing availability of data to offer a wide range of financial services, including but not limited to banking, payment processing, investment management, and insurance.[2]

Given their growing importance, new legislative frameworks (e.g., GDPR[3] and the CCPA[4]) have been introduced to regulate the collection and processing of personal data. Their main goal is to ensure that the ownership of information generated through consumers' digital activities is returned to the consumers themselves. In the context of financial markets, a more specific initiative known as *open banking* aims to "support innovation and competition in retail payments and enhance the security of payment transactions and the protection of consumer data." Open banking has the potential to reshape financial markets by enabling new entrants, such as fintechs, to compete with traditional financial

[1] Subsequent contributions have shown that under- or over-investment is possible. For instance, de Meza and Webb (1987) show that over-investment is a possible equilibrium outcome.

[2] Buchak et al. (2018) shows that the growth of fintech lending is driven by both regulatory arbitrage and technological innovation. Using mortgage data, they find that shadow banks, including fintechs, gained substantial market share from 2007 to 2015, with over half of the growth explained by lighter regulation and about a third by technology-enabled efficiency. For further discussion on the definition and evolution of fintechs, see Giglio (2021).

[3] General Data Protection Regulation; See `https://gdpr-info.eu/`.

[4] California Consumer Privacy Act; See `https://oag.ca.gov/privacy/ccpa`.

intermediaries in the provision of banking services.[5]

Open banking and the rise of new technologies that enhance the dissemination of information and improve predictive accuracy have the potential to redistribute market power and significantly reshape financial markets. This evolution raises several economically significant questions: under what conditions does the availability of more information, for instance through data-sharing initiatives such as open banking, benefit borrowers and lenders? When it does not, are there alternative institutional arrangements that lead to better market outcomes and improved welfare? And how does the prospect of better information shape lenders' incentives to improve predictive accuracy?

To address these questions, Section 2 presents a model of a credit market characterized by asymmetric information between lenders and borrowers. In particular, we consider an economy populated by financially constrained borrowers, a traditional bank, and a fintech company. Borrowers are endowed with risky projects that require a fixed capital investment to take off. The potential return of each project varies, with borrowers being differentiated based on the probability of generating a positive return. We categorize borrowers into two types: low and high, where only the high-type borrowers possess creditworthy projects. Additionally, borrowers have a fixed quantity of a physical asset at their disposal, which can serve as collateral should they opt for a loan.[6]

Our model features two key characteristics. First, we assume that lenders face a cost in collateralizing the physical asset. As in Bester (1985) and Boot et al. (1991),[7] we assume that due to transaction costs, legal expenses of foreclosure, political frictions, or other factors, the collateralized asset is worth more to borrowers than to lenders.[8] A pivotal assumption in our model, however, is that the bank holds a comparative advantage over

[5]Babina et al. (2025) shows that open banking indeed successfully promotes fintech entry. In particular, using a difference-in-differences design, they show that the number of VC-backed fintech financings increases by one-third and the total amount of capital invested doubles following the adoption of open banking policies.

[6]Collateral can take different forms. For example, short-term loans are sometimes collateralized by accounts receivable, in which case the lender seizes the borrower's cash flow if the borrower fails to fulfill their obligation. Some fintechs provide loans that are repaid directly through revenue collections. In these cases, the value of collateral to the lender is almost equal to that to the borrower. Given that a key feature of our model is that the value of collateral differs between lenders and borrowers and that the two lenders are differentiated, our framework applies more directly to settings involving physical collateral such as real estate, equipment, or inventory.

[7]One of the main differences between our paper and related papers of data sharing is the use of collateral as a screening device. For instance, He et al. (2023) does not allow for any screening device (such as collateral). Huang (2021) and Serfes et al. (2025) do allow for collateral but merely as an alternative mode of repayment. In our paper, collateral is a device that can be used to transmit information from the borrowers to the lenders.

[8]For instance, borrowers may derive greater utility from the asset because it is part of a productive enterprise or due to personal attachment (e.g., a primary residence). See Djankov et al. (2008), which estimates the combined costs associated with pledging and enforcing collateral across OECD countries.

the fintech in leveraging collateral as a means of securing loans. This advantage stems from the bank's institutional experience and legal infrastructure and is supported by empirical evidence.[9] Our model, therefore, applies better to loans to small firms (SME, SBA, etc.), as, for example, in Gambacorta et al. (2023), which considers SME loans obtained either by MYbank (a fintech) or traditional commercial banks.[10]

The second key component of our model is that the fintech is more capable than the bank in analyzing data from public sources or previous transactions and, hence, is able to make better predictions. In particular, we assume that the fintech can, with some probability (σ), identify whether a borrower is creditworthy. This probability measures the fintech's informational strength and market power. The advantage of fintechs over banks in analyzing data is well-documented (see the discussion in Section 3). To sum up, we study competition between a collateral-dependent bank and an information-reliant fintech.

Section 3 characterizes the unique mixed-strategy equilibrium (in non-dominated strategies).[11] We show that, in equilibrium, the bank randomizes over a set of secured loans, while the fintech randomizes over a set of unsecured loans. Recent empirical studies are, in fact, consistent with this equilibrium outcome (see the discussion in Section 3). In equilibrium, only high types borrow either by selecting a secured loan from the bank or an unsecured loan from the fintech.

Our main objective in this paper is to study the welfare consequences of strengthening the precision of the signal received by the fintech (σ). In other words, we are interested in how welfare is affected by improvements in information. Such improvements can stem from data transfers (e.g., open banking initiatives), enhancements in data analytics, or other technological developments. In Section 4, we study the welfare of the different market participants, as well as total welfare, as a function of the informativeness of the signal (σ).[12]

[9] Cerqueiro et al. (2020) examines an exogenous transfer of priority rights from banks to other creditors and shows that this shift affects future corporate financing, suggesting that banks have an advantage in utilizing collateral.

[10] In Gambacorta et al. (2023), collateral primarily refers to tangible assets—most notably real estate—used to secure bank loans. By contrast, big tech lenders offer mostly unsecured credit and instead rely on alternative data sources to assess creditworthiness.

[11] As discussed in the main part of the paper, our equilibrium shares features with the mixed-strategy equilibrium characterized in Blume (2003) and is reminiscent of the price dispersion model in Varian (1980).

[12] Among others, Thanassoulis and Vadasz (2025) studies how improvements in lenders' screening technologies interact with adverse selection and borrower naivety in competitive credit markets. A key difference from our paper is that we identify a novel channel through endogenous collateralization, showing that improved information can trigger destructive competition and reduce welfare. In contrast, their mechanism operates through price dispersion and naivety-driven profit incentives, abstracting from contract structure and collateral choices.

A critical feature of our equilibrium is that improvements in the fintech's signal precision reshape the market structure. An increase in σ strengthens the fintech's competitive position and compels the bank to respond with more attractive contracts. These contracts, however, carry higher collateral requirements. Better information thus has two opposing effects on welfare: it allows more borrowers to obtain non-collateralized loans, but it also intensifies competition, leading the bank to raise the collateral it demands. This trade-off underlies our welfare results.

As better information intensifies competition, borrowers benefit and enjoy higher welfare.[13] By contrast, a more powerful fintech reduces the bank's profit. For the fintech itself, the effect is more subtle: when the bank's value of collateral is low, the bank has limited capacity to respond, and the fintech's profit increases with σ. When the bank's collateral value is high, the relationship becomes non-monotonic: excessive competition can reduce the fintech's profit if σ is too high.

The effect on total welfare depends on the bank's value of collateral, which determines the strength of its competitive response to the fintech's improved information. When the bank's collateral value is low (its cost of collateralization is high), the response is limited, and the efficiency gains from borrowers moving into unsecured fintech loans prevail, so welfare rises with σ. When the bank is highly proficient at using collateral (its cost of collateralization is low), the response is strong enough that the resulting collateral inefficiency outweighs those gains, and total welfare can decline as σ increases.[14]

In Section 5, we study the implications of our results for open banking, the establishment of data markets (i.e., Laudon, 1996), and improvements in data analytics. For instance, recent empirical evidence (Babina et al., 2025; Nam, 2023) documents that open banking indeed reduces adverse selection, leads to an expansion of credit, and affects welfare.[15] We establish conditions under which open banking can enhance the welfare of market participants. We find, however, that open banking does not always improve total welfare and might even be harmful to fintechs. In situations where open banking may not be optimal, we show that market efficiency can sometimes be improved by re-

[13]If borrowers experience an intrinsic privacy cost (i.e., higher σ entails the use of more personal data) then borrower welfare may no longer be strictly increasing in σ and could instead exhibit an inverse-U shape. Nonetheless, all our main results remain robust.

[14]This stands in contrast to He et al. (2023), which demonstrates that open banking always increases market efficiency. Our paper complements their result in the sense that we provide conditions under which open banking can deteriorate total welfare while increasing borrowers' surplus.

[15]Nam (2023) analyzes loan-level data from a German fintech lender and finds that customer data sharing improves credit allocation (particularly for riskier borrowers) by increasing approval rates, lowering interest rates, and reducing defaults. Together, these studies underscore how data portability can reduce information frictions and expand financial access through more accurate screening and enhanced competition.

stricting the flow of data to fintechs to an optimal level below full sharing. These novel results contribute to the debate on open banking by highlighting that data sharing can have consequences opposite to those expected by policymakers.

We then consider alternative data-sharing institutions, such as the allocation of property rights and the establishment of a market for data. For instance, early work by Laudon (1996) envisioned the emergence of "National Information Markets," where individuals could sell their personal data. We find that while the allocation of property rights is sometimes irrelevant for welfare, in other cases, assigning property rights over data to lenders may lead to better outcomes. We derive implications for when this would be optimal. For example, we conjecture that in advanced economies with strong lender protection laws and judicial systems, open banking might be dominated by an alternative institution in which lenders are allocated the property rights and can trade in a well-established market for data.

Section 7 concludes the paper. All formal proofs are relegated to the Appendix. The Online Appendix considers several extensions of the model and establishes the robustness of our results under alternative specifications.

2 Model

Preliminaries

There are two time periods (today and tomorrow) and a single good. We will use the terms "good", "cash" and "dollars" interchangeably to represent the same thing. The net risk-free interest factor between these two periods is exogenously given and denoted by r. All agents in the economy are risk-neutral and do not discount the future. Consequently, all agents share the same preferences over (possibly random) consumption streams $(\tilde{c}_1, \tilde{c}_2)$ across the two periods, represented by

$$u(\tilde{c}_1, \tilde{c}_2) = \mathbb{E}[\tilde{c}_1] + \mathbb{E}[\tilde{c}_2],$$

where $\mathbb{E}[\cdot]$ denotes the expectation operator.

Borrowers

There is a continuum of borrowers of measure one. Borrowers can be of two types: high and low, indexed by $i \in \{H, L\}$. The share of type i borrowers in the population is $\lambda^i \in (0, 1)$, with $\lambda^L + \lambda^H = 1$. For notational simplicity, we let $\lambda^H \equiv \lambda$. The borrower's type (i) is (a priori) her private information.

Each borrower has a project that requires 1 dollar today to initiate and yields a random return $\tilde{Y}$ dollars tomorrow. The expected return of the project depends on the borrower's type. Specifically, the project yields $\tilde{Y} = Y$ with probability θ^i, and $\tilde{Y} = 0$ with probability $1 - \theta^i$, where $\Delta\theta \equiv \theta_H - \theta_L > 0$. Let $\theta^\circ \equiv \lambda^H\theta^H + \lambda^L\theta^L$ denote the average probability of success—this also corresponds to the fraction of borrowers whose projects succeed if all undertake them. We assume that the outcome of the project (i.e., the realized state) is observable and verifiable in a court of law. We further make the following assumption:

$$\frac{r}{\theta^H} < Y < \frac{r}{\theta^\circ}. \tag{1}$$

This assumption has several important implications, which we discuss below. First, since $\theta^L < \theta^\circ$, it implies that while high types possess projects with strictly positive net present value (NPV), low types possess projects with strictly negative NPV. Hence, low types should not undertake their projects, as doing so imposes a social cost.[16] Furthermore, the assumption implies that no lender would be willing to offer loans indiscriminately without additional information, because such indiscriminate lending would be unprofitable.

Borrowers are endowed with physical assets that can serve as collateral.[17] The collateral level is the same for all borrowers and is equal to $\bar{C}$. This means that if a borrower keeps her collateral until tomorrow, she can earn $\bar{C}$ in dollars. We assume that $\bar{C}$ is sufficiently high to enable effective screening.

Lenders

There are two types of lenders in our economy: a traditional bank and a fintech company, indexed by $\zeta = \{B, F\}$.[18] Each lender can raise funds inelastically at the risk-free interest rate (r) and can provide one-dollar loans to borrowers, where loans are in the form of secured debt.[19] Specifically, a loan contract is represented by a pair $x = (R, C)$, where R is

[16]Although not necessary for our results, this assumption simplifies the analysis. Allowing projects by low types to have positive NPV would imply that both types are granted loans in equilibrium, which is not the case under the information structure we consider below. See Section 6 and the Online Appendix for more details.

[17]It is rather straightforward to extend the model in a way that makes collateral productive. In particular, one can consider that collateral as physical capital, is combined with one unit of the good to hire labor, to produce output. What is important for our results is that when the collateral is transferred from the borrower to a lender, some value is lost. See below for details.

[18]Section 6 and the Online Appendix consider cases in which there are more banks and more fintechs and discusses in what sense our results are robust.

[19]Parlour et al. (2022) models competition between a fintech that provides payment services and a bank that competes in payment services but is a monopolist in the loan market. Instead, we consider competition between a fintech and a bank in the loan market.

the repayment amount (principal plus interest) a borrower owes to a lender, and C is the amount of collateral that the lender will seize in case the borrower defaults. Given our assumption that the outcome of the project is observable and verifiable, it is natural that contracts can be perfectly enforced ex post such that there is no strategic default.

Consistent with the assumption in Bester (1985, 1987), we posit that collateralizing assets incurs costs. These costs can be a combination of transaction costs (appraisal, legal, monitoring), risk costs (illiquidity, depreciation, foreclosure delays), and regulatory capital requirements. In particular, a lender needs to incur appraisal and legal costs before signing the contract to estimate and ensure the value of a collateralized asset. Moreover, in the event of a default, a lender might capture less than the true value of the collateral, potentially due to transaction costs or other inefficiencies involved in transferring the collateral from the borrower to the lender, which constitutes an implicit social cost.[20]

To provide an estimate, note that empirical evidence shows that the cost of collateralizing physical assets in OECD countries is nontrivial. Djankov et al. (2008) estimates that, across OECD jurisdictions, the combined costs associated with creating, registering, and enforcing collateral-including fees, legal expenses, administrative delays, and foreclosure-related inefficiencies-range from approximately 3% to 12% of the asset's real value.[21]

A key assumption in our analysis is that the cost of collateralization for fintechs is higher than for banks. This can be justified by positing that banks, thanks to their size and experience, have more efficient divisions managing foreclosures. Additionally, the lower collateralization cost for banks may be viewed as an advantage in lending to local firms by leveraging their established relationships. As highlighted in the introduction, this assumption is supported by empirical evidence. For example, Cerqueiro et al. (2020) demonstrates the collateralization advantage of banks compared to other creditors. Furthermore, studies such as Gambacorta et al. (2023) and Gopal and Schnabl (2022) underscore that fintechs rarely provide collateralized loans, which is also confirmed by the results of our paper.[22] Without loss of generality and to economize on notation, we

[20]There are several practical instances supporting this assumption. For example, consider a house as collateral. After foreclosure, a lender often incurs a significant loss in the house's value upon taking possession. Similarly, for a physical asset like a factory belonging to an established firm, the value can diminish considerably in foreclosure, either because the borrower has specialized knowledge in its utilization or due to unique applications of the asset that only the borrower can execute.

[21]To bring these insights into a more recent context, the World Bank's *Doing Business 2020* indicators for resolving insolvency in high-income OECD economies indicate that the direct cost of recovering debt measures on average around 9% of the debtor's estate. See World Bank, *Doing Business 2020: Resolving Insolvency. Available at:* https://archive.doingbusiness.org/content/dam/doingBusiness/media/Annual-Reports/English/DB12-Chapters/Resolving-Insolvency.pdf

[22]We show below that in equilibrium, fintechs offer unsecured loans whereas banks offer loans backed

assume that in the case where the bank provides a collateralized loan and seizes the collateral, it earns a (per unit of collateral) value $\gamma_B \equiv \gamma$, where $\gamma \in (0,1)$, whereas the fintech earns a (per unit of collateral) value $\gamma_F = 0$.[23]

The payoffs of type i from signing contract x and lender ζ from signing contract x with type i are given, respectively, by:

$$U^i(x) = \theta^i \max\{Y - R + \bar{C}, \bar{C}\} + (1 - \theta^i) \max\{\bar{C} - C, 0\} \tag{2}$$

and

$$\Pi^i_\zeta(x) = \theta^i R + (1 - \theta^i) \gamma_\zeta \min\{C, \bar{C}\}. \tag{3}$$

We assume that borrowers accept loan contracts with strictly positive expected payoffs, and they choose randomly if they are indifferent between two loan contracts.[24]

Information

Our goal is to study how information affects market outcomes and welfare. In particular, we assume that only the fintech receives a signal for each borrower who applies for a loan: $S \in \{S^+, S^-\}$. The assumption that fintechs obtain and rely on informational signals rather than on traditional lending channels is well-grounded. According to Giglio (2021), "fintech innovation in lending is found in the use of alternative credit models, online data sources, price risk data analysis, fast lending processes, and lower operating costs." Thus, the core of the fintech business model is built around predictive data analytics, further justifying our assumption that fintechs (and, to a lesser extent, traditional banks) receive informative signals.

The probability that the fintech receives signal S for a type-i borrower is denoted by $\sigma^i(S) \geq 0$, where $\sigma^i(S^+) + \sigma^i(S^-) = 1$.

In the main part of the paper, we focus on a *false-negatives (or partially inconclusive) signal structure*, which in our model corresponds to the assumption that $\sigma^H(S^+) \geq 0$ and $\sigma^L(S^+) = 0$ (See also Figure 1). In plain English, this means that the fintech may receive a negative signal even if the borrower is of high type, but can never receive a positive

by collateral.

[23] As we mentioned above, the assumption that the (per unit of collateral) value for the fintech is zero is without loss of generality. All our results hold as long as the per unit values of the two lenders satisfy the following inequality: $0 \leq \gamma_F < \gamma_B < 1$. See Section 6 and the Online Appendix.

[24] The modeling of competition between lenders with different costs of collateral has characteristics similar to Bertrand price competition with asymmetric marginal costs. As shown below, we follow Blume (2003) to find the mixed-strategy equilibria under the conventional demand sharing rule. We could replace this rationing rule by positing that, when indifferent between two loan contracts, borrowers choose the contract offered by the bank without changing the qualitative features of our results.

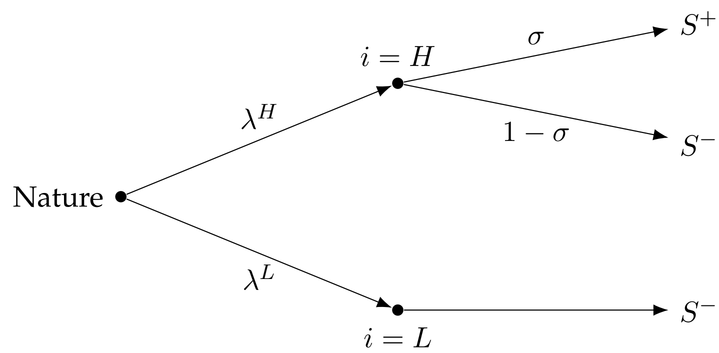


Figure 1: **The signal structure.** If the type is low, the fintech receives signal S^- with certainty; if the type is high, the fintech receives signal S^+ with probability σ and signal S^- with the complementary probability.

signal if the borrower is of low type. This immediately implies that upon receiving a positive signal, the fintech can perfectly identify that the borrower is of high type. Given this signal structure, and with some abuse of notation, we denote $\sigma^H(S^+) \equiv \sigma$. Although in the main part of the paper we assume that only the fintech receives a signal and is *partially inconclusive*, this assumption is by no means necessary for our results.[25]

Given the assumption that the average project has a strictly negative net present value (NPV), another implication of the assumed signal structure is that the fintech may offer an non-collateralized loan to a borrower upon receiving a positive signal; however, any such loan would necessarily yield a strictly negative profit if the signal is negative. Consequently, upon observing a negative signal, the fintech can only offer collateralized loans.

Timing of Events

The timing of events is specified below:

- **Stage 0**: Nature selects the type of every borrower and every borrower privately learns their type.

- **Stage 1**: The two lenders publicly offer signal-contingent loan contracts.[26] Borrowers observe the contracts offered by both lenders.

[25]In Section 6 and the Online Appendix, we discuss the robustness of our results in two extensions. First, in a model in which the bank also receives a signal. Second, in a model in which the fintech has a more general signal structure.

[26]In the baseline model, the bank has no signal and thus offers unconditional contracts; the fintech offers a menu of contracts—one for borrowers who receive a positive signal and one for those who receive a negative signal.

- **Stage 2**: Each borrower's signal from each lender is realized and privately observed by the borrower and the corresponding lender.

- **Stage 3**: Borrowers choose one of the two lenders or decide not to borrow. If a borrower chooses to borrow, the game proceeds to the next stage.

- **Stage 4**: The borrower receives funds under the contract corresponding to the offer made in *Stage 1* and the signal realized in *Stage 2*, and invests in the project.

- **Stage 5**: If the project succeeds, the borrower repays R; if it fails, the lender seizes the collateral C.

Under this timing, the two lenders publicly announce their menus, which are conditional on the borrowers' realized signals. Borrowers then observe their signals and choose whether to apply to one of the two lenders or to opt out and sign no contract. Once a contract is selected, the borrower invests in their project, and payments are executed based on the realized outcome.[27]

A key assumption in our model is that while each lender knows the signal structure of their competitor, they do not observe the signal that a borrower receives from the competing lender. Given our assumptions, this means that although the bank knows σ, it does not observe the signal of the fintech for any of the borrowers. Notably, the fintech has a strict incentive to treat its signals as proprietary information. This is because if the bank observed the signal received by the fintech, the two lenders would compete a la Bertrand, wiping out any positive profits. Given that, as we show below, in equilibrium, the fintech earns strictly positive profits for any informative signal, the fintech's profit strictly decreases if its signals are observed by the bank. This implies that the fintech strictly prefers to keep its signals private.

We analyze the perfect Bayesian equilibria (PBE) of the model. A strategy for a lender specifies a menu of contracts, contingent on the signal received by the borrower. A strategy for a borrower specifies, for each type and signal realization, whether to accept a contract from one of the lenders or to keep their collateral and abstain from borrowing. A PBE consists of strategy profiles for all lenders and borrowers such that: (i) each lender maximizes its expected profit given the strategies of the borrowers and the other lender; and (ii) each borrower maximizes their expected utility given the strategies of the lenders.

[27] A game that yields qualitatively similar results is one in which, in the first stage, lenders privately observe signals about each borrower and offer personalized (private) loan contracts. In the second stage, borrowers choose one of those contracts or opt out.

3 Equilibrium

In this section, we study the equilibria of the game. Figure 2 will significantly facilitate our exposition. The figure depicts the contract space, along with the indifference curves of the two types and the zero-profit lines of the two lenders. The steep (red) line originating at Y represents the set of contracts for which the low type is indifferent between applying for a loan and not borrowing:

$$\theta^L(Y - R) - (1 - \theta^L)C + \bar{C} = \bar{C} \tag{IC_0^L}$$

The less steep (green) line starting at c is an indifference curve for the high type:

$$\theta^H(Y - R) - (1 - \theta^H)C + \bar{C} = \bar{U}^H. \tag{IC^H}$$

The horizontal line beginning at r/θ^H and the steeper black line—also starting at r/θ^H–are the fintech's and bank's zero-profit lines:[28]

$$\theta^H R + (1 - \theta^H)\gamma_\varsigma C - r = 0. \tag{ZP_ς^H}$$

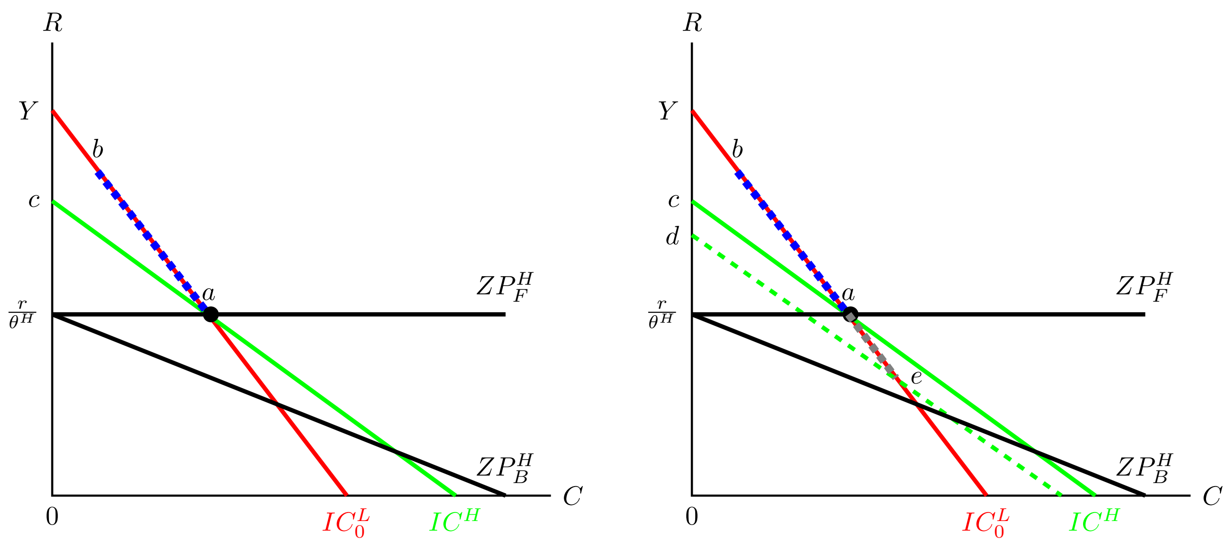


Figure 2: **A graphical representation of the equilibrium.**

The easiest and most pedagogical way to understand the equilibrium is to begin with a hypothetical scenario in which both lenders face the same cost of collateralizing assets, and the fintech has a completely uninformative signal, as in the seminal paper of Bester

[28] Recall that the fintech assigns zero value to collateral; hence, its profit lines are horizontal.

(1985). Recall that, under Assumption 1, any lender would like to exclude all the low types. Therefore, the lenders would like to offer a contract on the right of line (IC_0^L). Due to competition among lenders, the unique equilibrium entails both lenders offering a single collateralized contract that targets the high types, as depicted by contract a in Figure 2a.[29]

Now assume that the fintech's signal remains uninformative, but suppose that $0 = \gamma_F < \gamma = \gamma_B$. In this case, the bank has a clear advantage over the fintech due to its lower cost of collateralization. Ideally, the bank would like to offer the most profitable contract, but it is constrained by the competitive threat posed by the fintech. Because the two lenders are asymmetric, a pure strategy equilibrium does not exist.[30] For any pair of contracts offered by the two lenders, there always exists a profitable deviation, as in the classic Bertrand game with asymmetric sellers.

The unique mixed-strategy equilibrium (in non-dominated strategies) is illustrated in Figure 2a. In this equilibrium, the bank offers contract a, which is selected by the high types, whilst all low types refrain from borrowing. The fintech acts as a competitive threat, randomizing among the contracts that belong to (b, a) with full support. Neither lender has an incentive to deviate. The fintech randomized offer is such that the bank is weakly worse off offering any alternative contract that improves per-borrower profits relative to a, because its offer will be accepted with a lower probability. In this equilibrium, the fintech does not attract any borrower but has no profitable deviation: any contract better than a will necessarily yield losses.

Now suppose that the fintech's signal is informative (i.e., $\sigma > 0$). The fintech is able to identify a share of high-type borrowers; this allows it to attract them away from the bank by offering a contract slightly below point c in Figure 2a. The bank, in turn, will attempt to respond by offering a contract on the IC_0^L line that attracts all borrowers. This dynamic implies that a pure strategy equilibrium cannot be sustained. Unlike the case where $\sigma = 0$, however, the bank now randomizes over collateralized contracts.

Point e in Figure 2b represents the contract that, if accepted by all high types, yields the same profit to the bank as contract a would yield if taken only by the share of high types

[29]Note that this is essentially a variant of the Rothschild and Stiglitz (1976) least-cost separating equilibrium in which the two lenders earn zero profits, the high type accepts a collateralized loan, and the low type stays out. Assumption (1) ensures that no profitable deviation exists because every contract that attracts both types will necessarily yield strictly negative profits.

[30]The nature of the equilibrium in this setting shares key features with the classic Bertrand duopoly, in which two asymmetric firms with linear costs (a low-cost and a high-cost firm) compete by posting prices for a homogeneous product. Under the standard market-sharing rule, according to which the firm with the lower price captures the entire market and equal prices lead to a split of demand, it is well known that a pure strategy equilibrium fails to exist. See Blume (2003) and the remarks after Proposition 1.

who receive a negative signal from the fintech (i.e., $1 - \sigma$). In other words, the bank is indifferent between attracting all high types at contract e or attracting only the $1-\sigma$ share at contract a. Suppose now that the bank plays contract a with some non-zero probability (i.e., there is a mass point at a) such that the fintech is indifferent across the entire segment $(c, d]$. That is, the fintech is indifferent between offering a contract that attracts only the high types with a positive signal at point c and offering a contract that attracts all such types at point d.

We can fully characterize probability distributions for the two lenders such that the bank randomizes over contracts in the segment $[a, e]$, placing a mass point at a, while the fintech offers contracts in the segment $(c, d]$ to borrowers with a positive signal and contracts in $[b, a]$ to those with a negative signal such that no profitable deviation exists. As in the $\sigma = 0$ case, the bank is weakly worse off offering any contract above a (since those offered by the fintech act as a credible threat) or below a (as this would increase the share of high types attracted but reduce per-unit profits). Likewise, the fintech cannot profitably offer contracts above c (i.e., any such borrowers would instead choose the bank) or below d, which would increase the borrower share but reduce per-unit profits.

The following proposition summarizes the unique equilibrium in undominated strategies in this game.

Proposition 1. *There exists a unique class of mixed-strategy equilibria in undominated strategies in which the bank offers randomized loans backed by collateral, whereas the fintech offers randomized loans backed by collateral to borrowers with the negative signal and non-collateralized loans to borrowers with the positive signal. In equilibrium, only the high-type borrowers borrow either from the bank with collateral or from the fintech without collateral. All lenders earn positive profit for $\sigma \in (0, 1)$.*

Remark 1. In the classic model of Bertrand competition with asymmetric firms, Blume (2003) shows that a mixed-strategy equilibrium exists in which the low-cost firm sets a price equal to the high-cost firm's marginal cost, while the high-cost firm randomizes over higher prices. In other words, the high-cost firm acts as a competitive threat to the low-cost firm. Blume (2003) further shows that apart from this equilibrium, alternative equilibria exist that involve dominated strategies. In these equilibria, the low-cost firm selects a price below the marginal cost of the high-cost firm randomizes over a set of higher prices, thereby acting as a competitive threat.

Similar equilibria arise in our model. In particular, one can construct equilibria in which the bank randomizes over less-profitable contracts. As in Blume (2003), these equilibria involve dominated strategies. For this reason, we focus on equilibria in undominated strategies, as formally characterized in Proposition 1. The equilibrium we focus on is not only in undominated strategies but is also the unique equilibrium that yields the highest social welfare because it entails the least possible collateral.

Remark 2. It is worth noting that our model shares key features with the seminal paper by Varian (1980). In Varian's model, two sellers of a homogeneous good face two groups of buyers: informed buyers, who are aware of both sellers, and uninformed (or captive) buyers, who are aware of only one. Each seller aims to attract as many informed buyers as possible without sacrificing their captive buyers. This creates a trade-off: attracting more informed buyers requires lowering prices, which in turn reduces the profit extracted from uninformed buyers. The resulting equilibrium is in mixed strategies, with each seller randomizing over a set of prices in such a way that they earn exactly the profit they would obtain by selling only to their uninformed customers.

A similar mechanism arises in our setting. Since the bank has a comparative advantage in collateralizing assets, any borrower with a negative signal can be viewed as "uninformed" from the bank's perspective. In particular, the bank can profitably offer contract a (as shown in Figure 2) to such borrowers. Conversely, borrowers who receive a positive signal from the fintech can be seen as "informed". The equilibrium is again in mixed strategies: each lender randomizes over a set of contracts such that the bank is indifferent across offers and earns a profit equal to what it would obtain by attracting only the uninformed borrowers at contract a.

Remark 3. Our equilibrium is consistent with empirical evidence. For instance, Benmelech et al. (2024) documents that around 60% of firms with commercial bank loans in the U.S. have those loans secured and Berger and Udell (1990) finds that banks are more likely to demand collateral from riskier or less established borrowers, highlighting its role in mitigating information asymmetries. In contrast, recent studies such as Gambacorta et al. (2023) and Gopal and Schnabl (2022) show that fintech lenders rarely require collateral.

4 Welfare Effects of Information

Our primary interest in this paper lies in understanding how improvements in information affect equilibrium outcomes and welfare. Given our information structure, it is relatively straightforward that a higher σ is associated with "better information." Indeed, a higher σ implies that the fintech can more accurately identify the type of any borrower it interacts with; for instance, if $\sigma = 1$, the fintech is perfectly informed.[31]

Let the borrowers' expected payoff be denoted by $U(\sigma)$, and the payoffs of the bank and the fintech by $\Pi_B(\sigma)$ and $\Pi_F(\sigma)$, respectively.[32] Total welfare is then given by:

$$W(\sigma) = U(\sigma) + \Pi_B(\sigma) + \Pi_F(\sigma),$$

which corresponds to the welfare of a utilitarian social planner.

We begin by examining how a change in σ affects borrowers; in other words, we are interested in the shape of $U(\sigma)$. Recall that σ denotes the probability that the fintech correctly identifies a high-type borrower. As expected, an increase in σ allows the fintech to identify a larger share of high types and thereby expand its effective demand. In response, the bank competes by offering more attractive contracts to those borrowers who are identified by the fintech. This intensifies competition between the two lenders, each seeking to attract the high-type borrowers. As a result, an increase in σ strictly improves the payoff of the high types.

We now examine the effect of σ on the profits of the two lenders. First, note that the effect of σ on the bank's profit is unambiguously negative. A higher σ implies that the fintech can make more attractive offers and thus capture a larger share of the market at the bank's expense. Therefore, $\Pi_B(\sigma)$ is strictly decreasing in σ.

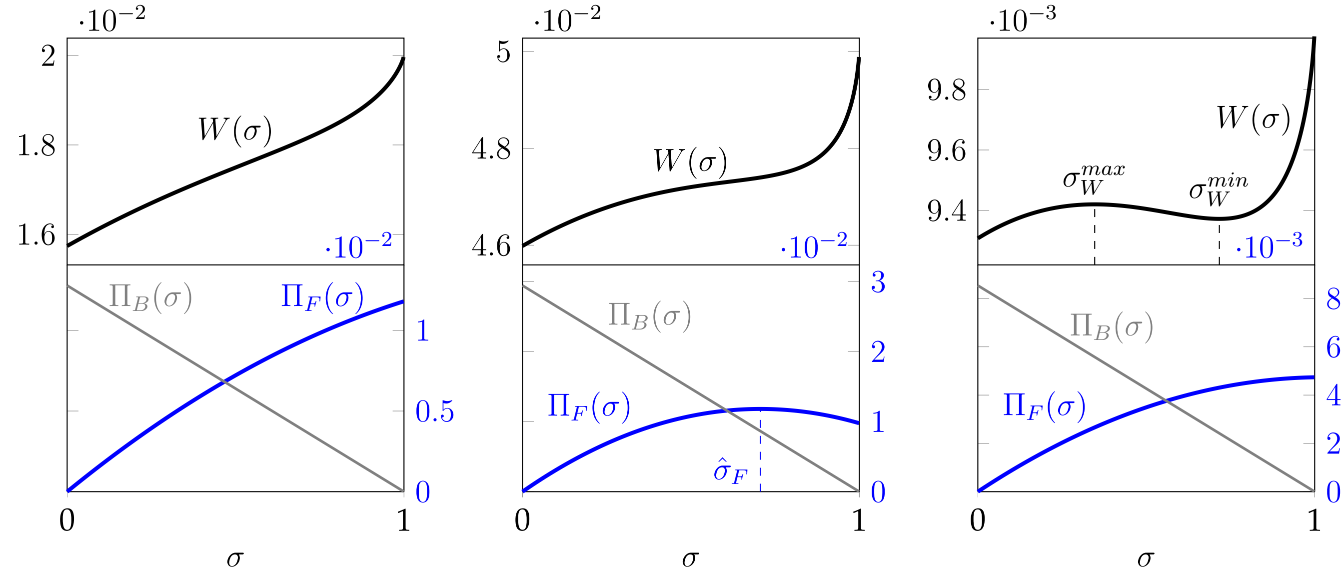
For the fintech, the effect of changes in σ is more nuanced. There are two competing forces at play. First, the *direct effect*: an increase in σ enables the fintech to identify more high-type borrowers and, thereby, expand its market share. Second, there is an *indirect effect*: as σ increases, per-customer profit decreases because the bank responds to the heightened competition by offering more attractive contracts, which forces the fintech to improve its own offers as well.

For sufficiently low values of σ, the direct effect dominates: increases in σ improve the fintech's position without substantially eroding per-unit profits. However, for higher values of σ, the fintech's profit becomes increasingly sensitive to the bank's counteroffers. The overall effect on $\Pi_F(\sigma)$ then depends critically on the parameter γ, which captures the bank's per-unit value of collateral.

Recall that a lower γ implies that the bank finds it more difficult to compete through collateralized contracts, which reduces its market power. In this case, the fintech enjoys

[31] Note that a higher σ implies that, conditional on the borrower being of high type, the probability of being correctly identified as high increases; similarly, if the borrower is of low type, the conditional probability of being identified as low also increases.

[32] We consider the payoff of borrowers before they learn their types (i.e., the *ex ante* payoff). Given that, for any parameter values, only the high types borrow, the payoff of the low types does not depend on σ. Therefore, the *ex post* payoff of the high types is $U(\sigma) - \lambda_B \bar{C}$.



(a) $\gamma < \min\{\gamma_F, \gamma_W\}$

(b) $\gamma_F < \gamma < \gamma_W$

(c) $\gamma_W < \gamma < \gamma_F$

Figure 3: **Total Welfare and lenders' profits as a function of the fintech's signal quality (σ).** (a) $Y = 2$, $\theta^H = 0.52$, $\theta^L = 0.48$, $\lambda = 0.5$, $\gamma = 0.75$. (b) $Y = 2$, $\theta^H = 0.55$, $\theta^L = 0.45$, $\lambda = 0.5$, $\gamma = 0.88$. (c) $Y = 2$, $\theta^H = 0.51$, $\theta^L = 0.49$, $\lambda = 0.5$, $\gamma = 0.925$.

a competitive advantage, and its profit $\Pi_F(\sigma)$ is strictly increasing in σ. As γ increases, however, the bank is better able to respond with more attractive collateral-based offers, thereby regaining market share. This diminishes the fintech's competitive edge. Hence, there exists a critical threshold γ_F above which the fintech's profit becomes inverse-U shaped in σ: initially increasing due to the direct effect, but eventually decreasing as the indirect competitive pressure dominates.

We can equivalently examine the effect of σ on total welfare. As noted earlier, borrowers always benefit from increases in σ, while the bank's payoff strictly decreases. The effect on the fintech, however, is ambiguous. From a welfare perspective, two forces are at play. The first is a *direct effect*: as σ rises, a greater share of borrowers obtain non-collateralized loans, thereby avoiding costly collateral requirements. The second is an *indirect effect*: higher values of σ intensify competition, forcing the bank to respond by offering contracts with higher collateral requirements. When γ is low, the direct effect dominates, so welfare increases with σ. By contrast, when γ is high, the indirect effect may dominate, and welfare can decline as σ rises. Hence, there exists a threshold value γ_W such that for $\gamma \le \gamma_W$, welfare is strictly increasing in σ, while for $\gamma > \gamma_W$, the relationship between welfare and σ may turn negative.[33]

[33] The threshold values γ_F and γ_W are not necessarily comparable; they are context-dependent. Depending on the parameter configuration, γ_F can be either lower or higher than γ_W. Examples are presented in

Proposition 2. *The following results hold regarding the effect of σ on payoffs and welfare:*

1. ***(Borrowers)*** *The payoff of the borrowers, $U(\sigma)$, is continuous and strictly increasing in σ.*
2. ***(Lenders)*** *The bank's profit, $\Pi_B(\sigma)$, is strictly decreasing in σ. Moreover, there exists a threshold $\gamma_F \in (0,1)$ such that:*
   (a) *For $\gamma \leq \gamma_F$, the fintech's profit $\Pi_F(\sigma)$ is strictly increasing in σ.*
   (b) *For $\gamma > \gamma_F$, the fintech's profit $\Pi_F(\sigma)$ has an inverse-U shape in σ.*
3. ***(Total Welfare)*** *There exists a threshold $\gamma_W \in (0,1)$ such that:*
   (a) *For $\gamma \leq \gamma_W$, total welfare $W(\sigma)$ is strictly increasing in σ.*
   (b) *For $\gamma > \gamma_W$, total welfare $W(\sigma)$ is bimodal, exhibiting an interior local minimum and an interior local maximum in $\sigma \in (0,1)$.*

Figure 3 represents different cases for the profit and total welfare functions depending on the parameters. For instance, in Figures 3a and 3b, total welfare is strictly increasing in σ, whereas in Figure 3c, total welfare is bimodal and exhibits a local maximum (σ_W^{max}) and a local minimum (σ_W^{min}). In Figures 3a and 3c, the profit of fintech is strictly increasing in σ, whereas in Figure 3b, the profit has an inverse-U shape, attaining an interior global maximum at ($\hat{\sigma}_F$).

In the following section, we discuss the implications of these results in various settings.

5 Implications

We now derive several implications from the results established in the previous section. To do so, we assume that $\sigma \in [\underline{\sigma}, \overline{\sigma}]$, where $0 \leq \underline{\sigma} < \overline{\sigma} < 1$. This reflects the fact that the informativeness of the signal may vary depending on the institutional environment (as discussed below). To understand how different levels of informativeness arise, we must address two important and interrelated questions: (i) why fintechs (and, more generally, lenders) are able to acquire information about potential borrowers, and (ii) how the prediction of borrower creditworthiness is improved.

The primary source of information for lenders is data analytics. This data may be publicly available (e.g., information shared on social networks, cookies, and other digital traces) or proprietary, including detailed financial records such as a borrower's wealth,

Figures 3b and 3c.

assets, credit history, and digital footprints. For instance, a lender may possess confidential information about specific borrowers, either from prior lending relationships or through external data sources.[34] Analyzing such data enables lenders to partially infer a borrower's type. We therefore interpret $\underline{\sigma}$ as reflecting the fintech's baseline technology (e.g., access to public data or social media), while $\overline{\sigma}$ corresponds to the highest achievable predictive accuracy given existing technological and data-processing capabilities.

Improvements in information quality (i.e., increases in σ) can arise from two main sources: (i) expanded access to borrower data (e.g., through open banking initiatives or data-sharing arrangements), and (ii) advances in data analytics that enhance predictive accuracy. That is, predictions improve either because of higher data input into a given technology or improvements in the technology (e.g., improvements in algorithms) itself. According to empirical evidence, lenders can significantly improve their prediction of borrower creditworthiness by leveraging digital footprints, alternative data sources, and machine learning techniques (Berg et al., 2020; Fuster et al., 2022).

In what follows, we examine the different possibilities in detail.

5.1 Data Transfers: Implications for Open Banking and Data Markets

5.1.1 Open Banking

The adoption of the *(Revised) Payment Services Directive* (known as PSD2) provided the legal framework governing data ownership and sharing within the European Union.[35] PSD2 requires European banks to offer automated access to customer transaction accounts, subject to customer consent, to qualified third parties via API technology.[36] This initiative, commonly referred to as *open banking*, aims to "support innovation and competition in retail payments and enhance the security of payment transactions and the protection of consumer data."

Under open banking, borrowers hold full property rights over their generated data and may transfer these data between lenders at will. Suppose, for instance, that borrowers have accumulated transaction (legacy) data from previous interactions with the bank, which they can choose to transfer to the fintech. The fintech may then use these data to

[34] Examples include soft information accumulated through lending relationships (Petersen and Rajan, 1994) and access to shared credit histories via credit bureaus (Pagano and Jappelli, 1993).

[35] For further information, see `https://www.ecb.europa.eu/paym/intro/mip-online/2018/html/1803_revisedpsd.en.html`.

[36] This access applies to both retail and corporate customers. APIs enable synchronization and interconnection of databases. Within the banking system, APIs link a bank's database (which stores customer information) with various applications and services, allowing the delivery of personalized products and payment solutions. See Barba Navaretti et al. (2017), pp. 43–46.

improve its prediction of borrower type. Naturally, more data enable better prediction, in the sense of a higher value of σ. At one extreme, no data sharing (closed banking) results in the lowest possible signal precision, $\underline{\sigma}$; at the other, full data sharing (open banking) yields the highest possible prediction, $\overline{\sigma}$.

Recall from Proposition 2 that the borrowers' payoff $U(\sigma)$ is strictly increasing in σ. It follows that borrowers strictly prefer to transfer their data to the fintech, assuming that doing so improves prediction. To keep the analysis tractable, we do not formally model the data transfer decision. Instead, we assume that under open banking, the fintech has access to the borrower's data before making any offer.[37]

Regarding lender profits, Proposition 2 shows that the bank is always worse off as σ increases. Therefore, open banking unequivocally reduces the bank's profit. For the fintech, however, the effect is ambiguous. As shown in Proposition 2, when γ is sufficiently low, the fintech's profit $\Pi_F(\sigma)$ is strictly increasing in σ, whereas for high γ, has an inverse U-shape. Thus, open banking may either benefit or hurt the fintech. Let

$$\hat{\sigma}_F \in \underset{\sigma \in [\underline{\sigma}, \overline{\sigma}]}{arg\,max}\, \Pi_F(\sigma) \tag{4}$$

denote the signal that maximizes the fintech's profit.

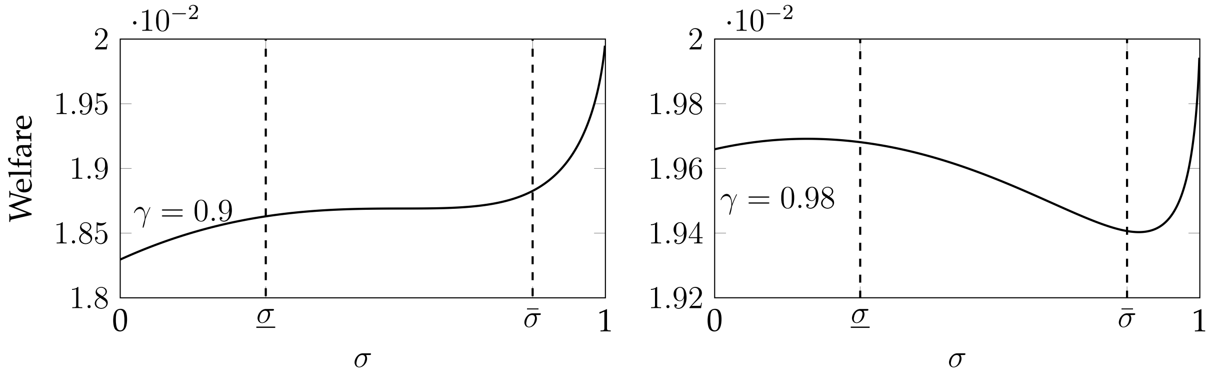
At one extreme, for $\hat{\sigma}_F = \overline{\sigma}$, open banking benefits the fintech as its profit is strictly increasing in the amount of data it receives. At the other extreme, for $\hat{\sigma}_F = \underline{\sigma}$, the fintech would strictly prefer closed banking. In any other case, *open banking with partial data sharing* may yield better outcomes than full data sharing. Specifically, suppose that $\hat{\sigma}_F \in (\underline{\sigma}, \overline{\sigma})$, that is, the fintech's profit function $\Pi_F(\sigma)$ is inverse- shaped.[38] Then, the fintech would strictly prefer to have access to only a subset of the data. Since more data increase σ, and the optimal point is interior, partial data sharing would be optimal from the fintech's perspective.[39]

A similar logic applies to total welfare. Recall from Proposition 2 that $W(\sigma)$ is either strictly increasing or bimodal, with both a local minimum and a local maximum in the

[37]This assumption corresponds to an equilibrium outcome. Suppose, for example, there is a preliminary stage (prior to Stage 0) in which borrowers decide whether to allow the fintech access to their data. In the continuation game, borrowers anticipate that a higher σ increases their expected payoff, and thus choose to share the data. Even if the decision were made after types are realized, a standard unraveling argument implies that both types would eventually agree to share their data.

[38]Reconsider Figure 3c and suppose that $\underline{\sigma} < \hat{\sigma}_F < \overline{\sigma}$.

[39]It is worth mentioning that several jurisdictions already implement forms of restricted data access in open banking. For example, the EU's PSD2 framework allows banks to provide access through tiered APIs with different scopes of data (e.g., account information only vs. payment initiation). Similarly, some platforms use throttling, rate limits, or require purpose-specific consent to control the flow and granularity of data.



(a) $\gamma < \gamma_W$ (b) $\gamma > \gamma_W$

Figure 4: **Total Welfare as a function of the fintech's signal quality**. $Y = 2$, $\theta^H = 0.52$, $\theta^L = 0.48$, $\lambda = 0.5$, $\underline{\sigma} = 0.3$, $\bar{\sigma} = 0.85$

interior. Let

$$\hat{\sigma}_W \in \underset{\sigma \in [\underline{\sigma}, \overline{\sigma}]}{arg\,max}\, W(\sigma) \tag{5}$$

denote the signal that maximizes total welfare. Then, the same conclusions follow: full data sharing (i.e., open banking with $\sigma = \overline{\sigma}$) may or may not be welfare-maximizing, depending on whether $\hat{\sigma}_W = \overline{\sigma}$ or lies in the interior of the interval.

Proposition 3 (Open Banking and Welfare)**.** *The welfare and distributional effects of open banking depend critically on the informativeness of the signal and the value of collateral:*

1. ***(Borrowers)*** *borrowers always benefit from open banking with full data sharing.*

2. ***(Bank)*** *The bank strictly prefers closed banking.*

3. ***(Fintech)*** *If $\hat{\sigma}_F = \overline{\sigma}$, the fintech prefers open banking with full data sharing. In all other cases, it may prefer either partial data sharing or closed banking.*

4. ***(Welfare)*** *If $\hat{\sigma}_W = \overline{\sigma}$, total welfare is maximized under full data sharing. Otherwise, partial data sharing or closed banking may be preferred.*

Recall that both $\hat{\sigma}_F$ and $\hat{\sigma}_W$ can lie in the interior of $[\underline{\sigma}, \overline{\sigma}]$ only if γ is sufficiently high. This leads to the following result:

Corollary 1. *If γ is sufficiently low, open banking (with full data sharing) is always welfare-enhancing; if γ is high, closed banking or open banking with partial data sharing might be socially optimal.*

Figure 4 depicts different cases in which open banking may or may not be welfare-enhancing. For example, in 4a, γ is low; hence, the total welfare is strictly increasing in the informativeness of the signal. In this case, open banking with full data sharing would be optimal. In Figure 4b, γ is high; hence, total welfare is non-monotonic. In this case, open banking with full data sharing is not optimal. Partial data sharing is optimal, but even closed banking dominates open banking.

This result allows us to derive important welfare implications regarding the effects of open banking. In particular, to the extent that γ reflects institutional characteristics, such as the level of economic development or the advancement and depth of a country's banking and financial systems, Proposition 3 implies that open banking is almost certainly welfare-enhancing in underdeveloped or emerging markets, where γ is likely to be low.[40]

Even in advanced economies, open banking is more likely to yield positive welfare effects when the effective value of collateral (γ) remains relatively low, such as during the early stages of fintech diffusion.

5.1.2 Property Rights and Data Markets

A common and important approach in economics is to compare the allocation of property rights among different participants and the creation of markets. For instance, the celebrated Coase Theorem (Coase 1960) establishes that when parties can bargain without frictions, they can not only reach mutually beneficial agreements but even agreements that are Pareto efficient. More importantly, and strikingly, Coase (1960) argues that the allocation of property rights affects only the distribution of surplus but not total welfare. Subsequent contributions challenged these conclusions. Among the most influential were those by Williamson (1985), Grossman and Hart (1986), and Hart and Moore (1990), who showed that when parties make non-contractible ex ante investments and lack commitment power, the allocation of property rights does matter. In such environments, ownership determines incentives and affects both investment and welfare outcomes.

Open banking, as studied in the previous section, can be interpreted as allocating full property rights over data to one side of the market: the borrowers. Under open banking, borrowers control the data and may consent to its transfer to the fintech. In line with the incomplete contracts literature (Williamson, 1985, Grossman and Hart, 1986, and Hart

[40] The value of collateral, γ, depends, apart from its physical or financial characteristics of the asset, on the institutional environment that governs enforcement. In underdeveloped economies, legal frictions, such as slow judicial processes, weak property rights, and high enforcement costs, significantly reduce the lender's ability to seize and liquidate collateral. As shown by Benmelech et al. (2024), reductions in the enforceability of collateral—due to legal changes or institutional inefficiencies—lead to credit contraction and macroeconomic slowdowns, highlighting that collateral value is endogenous to the legal system.

and Moore, 1990) and assuming that contracts cannot specify ex ante how much data must be shared, Proposition 3 characterized the conditions under which full data sharing, partial data sharing, or closed banking is optimal for the different participants.

We now extend this reasoning by comparing different allocations of property rights. We consider two alternatives: (i) borrowers own the rights (as under open banking), or (ii) one of the two lenders owns them.[41] If borrowers own the data and under the no-commitment assumption, the result mirrors open banking: they will voluntarily share all data with the fintech.[42]

An alternative is to assign property rights to one of the lenders. In this case, suppose that there exists a market for data, similar to that envisioned in Laudon (1996) and elaborated by more recent studies.[43] To keep the model tractable, we assume that the two lenders can negotiate over data usage in a way that maximizes their joint profits. When either lender owns the data, they will agree on the level of informativeness that maximizes their total surplus. Formally, the chosen signal is given by:

$$\hat{\sigma}_{BF} \in \underset{\sigma \in [\underline{\sigma}, \overline{\sigma}]}{arg\,max}\, \Pi_B(\sigma) + \Pi_F(\sigma). \tag{6}$$

Recall that $\Pi_B(\sigma)$ is strictly decreasing in σ, whilst $\Pi_F(\sigma)$ is either strictly increasing for low γ and inverse-U shaped for high γ. The sum of the two profits can be either strictly increasing for low γ or inverse U-shaped for high γ as exemplified in Figure 5.

The following lemma provides a complete characterization of the maximizer of the joint profit.

Lemma 1. *There exists $\gamma_{BF} < \min\{\gamma_F, \gamma_W\}$ such that*

1. *For $\gamma \leq \gamma_{BF}$, $\Pi_B(\sigma) + \Pi_F(\sigma)$ is strictly increasing in σ.*

2. *For $\gamma > \gamma_{BF}$, $\Pi_B(\sigma) + \Pi_F(\sigma)$, inverse-U shaped in σ.*

[41] Jones and Tonetti (2020) and Dosis and Sand-Zantman (2024) also study the allocation of property rights over online-generated data. Our paper relates to this literature by showing that, when a market for data exists, the allocation of property rights can influence welfare outcomes. Unlike Dosis and Sand-Zantman (2024), we consider multiple sellers (i.e., lenders) and allow these to trade property rights.

[42] This remains true even if there exists a market for data. So long as the data increase both the borrowers' and the fintech's payoffs, a mutually beneficial transaction will take place. A price for data will be negotiated such that the fintech gains access to the full dataset.

[43] A growing literature explores the idea of data as a tradable asset and the institutional design of data markets. Laudon (1996) proposes formal markets where individuals can sell their personal information under clearly defined property rights. Varian (1997) analyzes the economic characteristics of information goods, including pricing strategies and licensing. Acquisti et al. (2016) surveys the economics of privacy, emphasizing trade-offs between efficiency, disclosure, and regulation. Spiekermann et al. (2015) examines the ethical and institutional challenges of personal data markets.

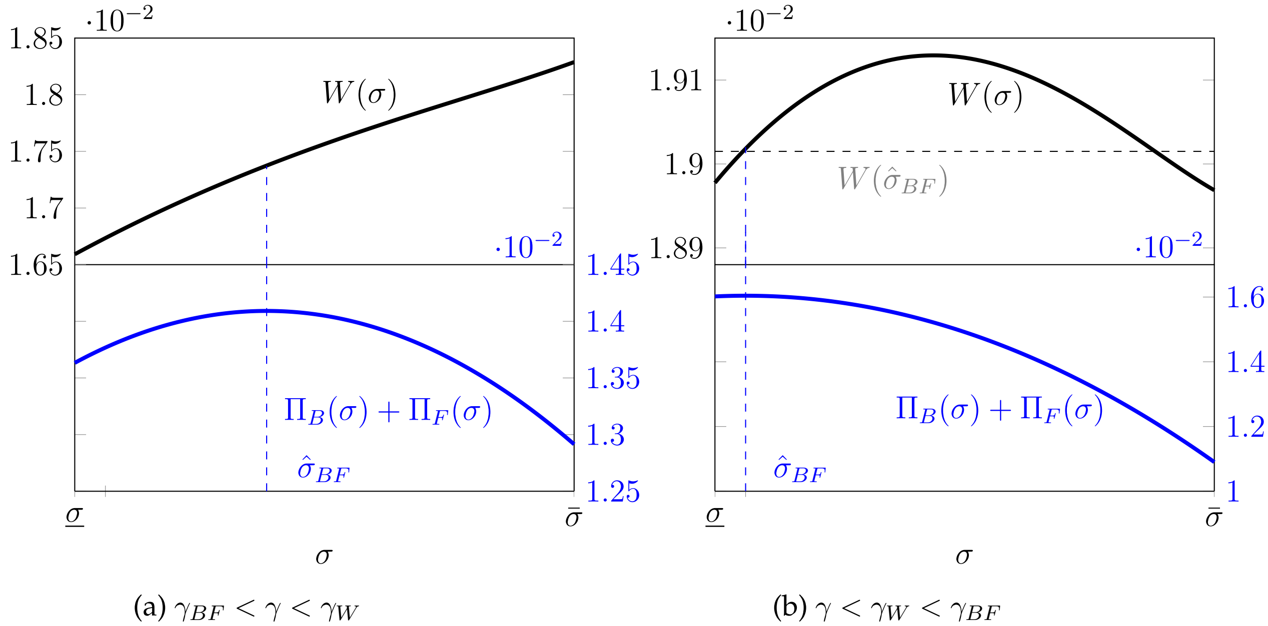


Figure 5: **Welfare and joint profit as a function of fintech's signal**. (a) $Y = 2$, $\theta^H = 0.52$, $\theta^L = 0.48$, $\lambda = 0.5$, $\gamma = 0.8$, $[\underline{\sigma}, \bar{\sigma}] = [0, 0.7]$. (b) $Y = 2$, $\theta^H = 0.55$, $\theta^L = 0.45$, $\lambda = 0.5$, $\gamma = 0.94$, $[\underline{\sigma}, \bar{\sigma}] = [0, 0.7]$.

We can now, based on Lemma 1, characterize the optimal allocation of property rights. In particular, when γ is sufficiently low, the resulting outcome is identical under any allocation of property rights; hence, the specific allocation is irrelevant for welfare. This equivalence, however, breaks down when γ is high. In that case, it is possible to have $\hat{\sigma}_{BF} < \overline{\sigma}$, in which case the allocation of property rights matters and can have significant implications. A full characterization is summarized in the following proposition.

Proposition 4 (Optimal Allocation of Property Rights)**.** *Suppose there is a functioning market for data. The optimal allocation of property rights depends critically on the informativeness of the signal and the value of collateral:*

1. *If $\gamma \leq \gamma_{BF}$, the allocation of property rights is irrelevant for welfare. In this case, open banking leads to the same outcome as assigning property rights to either of the two lenders.*

2. *If $\gamma_{BF} < \gamma < \gamma_W$, the allocation of property rights is either irrelevant or open banking is strictly preferable. In particular, when $\hat{\sigma}_{BF} < \overline{\sigma}$, open banking yields strictly higher welfare than any allocation in which a lender owns the rights.*

3. *If $\gamma \geq \gamma_W$, the optimal allocation of property rights is context-specific. If $W(\hat{\sigma}_{BF}) < W(\hat{\sigma}_W)$, open banking maximizes social welfare; otherwise, either the allocation of property rights is irrelevant, or allocating property rights to a lender yields superior outcomes.*

Recall that due to limited commitment, the allocation of property rights becomes welfare-relevant. When borrowers own the rights, they are willing to sell all their data to the fintech, knowing this improves their welfare.[44] On the other hand, when one of the lenders owns the rights, data sharing can be fine-tuned through negotiation. Specifically, the bank and the fintech have opposing preferences. A mutually beneficial agreement can therefore emerge, maximizing joint profits and leading to surplus-sharing. In this sense, assigning data ownership to one of the lenders serves as a partial commitment device, limiting σ and potentially improving welfare by constraining over-sharing.

In line with the previous analysis, Proposition 4 implies important policy conclusions regarding the optimal institutional design. Conditional on the existence of a data market, the allocation of property rights is irrelevant for social welfare when γ is low. For intermediate γ, either the allocation of property rights is irrelevant or open banking is strictly preferable. This suggests that in environments with relatively low γ, open banking may indeed be the welfare-maximizing institutional choice.

By contrast, when γ is high, open banking is not necessarily optimal. In such cases, it may be dominated by institutional frameworks in which data property rights are assigned to lenders, along with access to a well-structured market. These insights carry important policy implications for advanced economies, where high collateral value and institutional sophistication may call for more nuanced data governance models.

Note that several real-world initiatives seek to institutionalize data sharing through structured markets and alternative governance models. For instance, *Ocean Protocol* provides blockchain-based infrastructure for decentralized data marketplaces, using data NFTs, datatokens, and compute-to-data mechanisms to support data monetization while reducing the need to expose raw data. *Dawex* provides data exchange and data marketplace technology through which organizations can source, distribute, license, and monetize data products under configurable contractual and governance terms. Another emerging approach is that of *data unions* and related collective data intermediaries, which enable individuals or communities to pool data-related rights and negotiate the terms of data access, use, and benefit sharing collectively. These models broaden the logic of consent-based data access beyond open banking, aiming to better align data control, privacy, governance, and economic incentives.

[44] As discussed earlier, in the absence of commitment, the fintech will ultimately acquire all the data. This makes the outcome under borrower ownership equivalent to full open banking.

5.2 Improvements in Data Analytics and Predictions

Our framework also sheds light on the incentives of lenders, such as fintechs, to invest in acquiring and processing information.

From an economic perspective, it is important to compare the fintech's private incentives for information acquisition with what is socially optimal. In our model, this corresponds to comparing the fintech-optimal signal precision $\hat{\sigma}_F$, per equation (4), with the socially optimal level $\hat{\sigma}_W$, per equation (5). For simplicity, we assume that acquiring information is costless for the fintech.[45] The key question, then, is whether the fintech's privately optimal choice of σ aligns with the level that maximizes social welfare.

Recall that for $\gamma \leq \min\{\gamma_F, \gamma_W\}$, both the profit of the fintech and total welfare are strictly increasing in σ. In this case, the fintech would choose the highest possible level of informativeness (i.e., $\hat{\sigma}_F = \overline{\sigma}$), which also coincides with the socially optimal choice under a utilitarian planner maximizing total surplus. In other words, when γ is sufficiently low, the incentives of the fintech align with those of society.

For higher values of γ, several possibilities emerge. Suppose, for instance, that $\gamma_F < \gamma_W$ and $\gamma \in (\gamma_F, \gamma_W)$. In this case, $\hat{\sigma}_F$ may be interior while $\hat{\sigma}_W = \overline{\sigma}$, implying that the fintech underinvests relative to the social optimum. Conversely, if $\gamma_F > \gamma_W$ and $\gamma \in (\gamma_W, \gamma_F)$, the fintech overinvests. In short, when γ is sufficiently high, both over- and under-investment are likely depending on parameter values. This leads to the following result:

Proposition 5 (Improvements in Data Analytics)**.** *Suppose that the fintech can determine the informativeness of the signal through investment in data analytics and/or information processing.*

1. *For $\gamma \leq \min\{\gamma_F, \gamma_W\}$, the fintech's incentives align with those of society.*
2. *For $\gamma > \min\{\gamma_F, \gamma_W\}$, misalignment arises, and both over- and under-investment are possible.*

Proposition 5 highlights the role of non-internalized externalities. The fintech's investment decision affects all other participants in the market (i.e., borrowers and the bank). When γ is low, both the fintech and society have strong incentives to increase σ to avoid the cost of collateralization, resulting in aligned objectives. When γ is high, however, this alignment breaks down, leading to either excessive or insufficient investment in data analytics.

[45] Introducing a convex cost $\psi(\sigma)$ would not alter the qualitative conclusions: since $\Pi_F(\sigma)$ is either strictly increasing or inverse-U shaped, the net payoff $\Pi_F(\sigma) - \psi(\sigma)$ remains concave, implying a unique interior optimum.

These insights yield important policy implications. When γ is low, such as in emerging markets with weak collateral enforcement or high formalization costs, fintechs already have strong private incentives to invest in superior data analytics, and regulatory intervention may be unnecessary or even distortive. By contrast, when γ is high, private incentives may not be aligned with those of a social planner. In such contexts, fintechs may either overinvest in data analytics to gain a competitive advantage or underinvest if excessive precision reduces profitability. This divergence justifies targeted policy tools such as investment tax credits, data-access subsidies, or regulation of predictive technologies to align private incentives with social welfare. For example, the European Commission's Digital Finance Package explicitly supports open data access and ethical AI in credit scoring, precisely to manage this trade-off between innovation and inclusion.[46]

6 Discussion

The baseline model is intentionally stylized to isolate the central result of the paper: better information reduces information asymmetries (positive effect) but intensifies competition and may induce excessive collateralization (negative effect). In the Online Appendix, we show that our results are robust in several extensions of the model.

Bank Data Analytics. We allow the bank, alongside the fintech, to receive an informative signal. This case is important because banks also use internal account histories, credit bureau data, and proprietary scoring models to estimate borrowers' risk. We show that when the fintech has a more informative signal than the bank, an improvement in the fintech's signal attracts a larger share of high-type borrowers, which induces the bank to respond with more competitive offers. In that sense, our main results continue to hold: the effects of improvements in information depend on the value of γ, similarly to the baseline model.[47]

We also consider improvements in the bank's information, either through data-sharing arrangements that ease the bank's access to borrowers' data or through investment in information technology. Better information reduces the bank's reliance on collateral to mitigate adverse selection and, as such, unambiguously enhances welfare by lowering the inefficiencies of collateral-based lending. Unlike an improvement in the fintech's infor-

[46]See European Commission (2020), *Digital Finance Strategy for the EU*, COM(2020)591. Available at: `https://finance.ec.europa.eu/system/files/2020-09/200924-digital-finance-strategy_en.pdf`.

[47]He et al. (2023) shows that when both lenders receive a signal, the lender with the weaker signal earns zero profits. Unlike He et al. (2023), we show that both lenders have positive profits irrespective of whether the fintech or the bank has a more informative signal.

mation, better information for the bank is always beneficial because it does not trigger excessive collateralization.

This distinction sharpens the paper's policy message. Data sharing is not inherently welfare enhancing or welfare reducing. Its effects depend on whose information set improves and how the improvement changes the equilibrium use of collateral.

Multiple Banks and Fintechs. Although we concentrate on a market with two lenders, one bank and one fintech, the mechanism survives as long as competition remains imperfect and some lender has a collateral advantage in a market segment. In particular, what is crucial for our results is that lenders retain some form of local market power–an assumption consistent with the evidence.[48]

We first consider a spatial model in which two banks compete over a set of borrowers.[49] The distance between a bank and a borrower determines the bank's cost of collateralization, where distance can reflect local expertise, relationship lending, or specialization, as well as physical location. A bank that is close to a borrower in this space therefore extracts more value from her assets, so that different banks hold a collateral advantage in different market segments. In each segment, improvements in the fintech's information trigger the same response as in the baseline model: the collateral-driven bank competes more aggressively by adjusting its secured contracts, and welfare can become non-monotonic in the fintech's information when that bank is sufficiently proficient in using collateral, that is, when its γ is high.

The overall welfare effect of the fintech's information then depends on the range of $\gamma(\cdot)$ and the distribution of borrowers across segments. Full data sharing can be suboptimal relative to partial data sharing or to allocating property rights to lenders in a data market when proficient banks closely cover most segments. By contrast, when banks do not closely cover all segments, so that many borrowers face a high collateral cost with every bank, full data sharing outperforms alternative arrangements.

We also consider multiple fintechs that receive imperfectly correlated private signals. In this case, we show that each fintech may retain an informational rent. Numerical simulations demonstrate that the qualitative welfare patterns from the baseline model carry

[48] Relationship lending and banks' local market power are well documented in the banking literature; see for instance Boot (2000) and Degryse and Ongena (2005).

[49] This extension relates to Vives and Ye (2024), who provide a spatial model of loan markets to study how information technology affects competition, stability, investment, and welfare. They show that an improvement in IT incentivizes investment, but its effect on competition, stability, and welfare depends on whether IT weakens the influence of lender-borrower distance on monitoring costs. Our results align with their insights on the adverse effects of intensified competition, but our model differs by emphasizing the technological differences between banks and fintechs in mitigating adverse selection.

over to this environment. In particular, borrower surplus may increase with fintech signal quality, while total welfare, fintech profits, and joint lender profits can be non-monotonic. Hence, the implications for open banking, partial data sharing, data markets, and private incentives to invest in data analytics are not artifacts of the duopoly benchmark.

General Signals. We enrich the signal structure by allowing the fintech to receive imperfect positive signals, so that a positive signal no longer perfectly identifies a high type. In this case, an additional effect appears that reinforces our main result, similar to the *winner's curse* in Broecker (1990) and more recently He et al. (2023). When the fintech offers an unsecured contract, every low-type borrower who receives a positive signal accepts it, while high-type borrowers choose between the fintech's offer and the bank's collateralized loan. The fintech thus attracts an adversely selected pool: the low types it wrongly clears all accept, whereas the high types may still be drawn away by the bank.

Improved fintech information then affects welfare through three channels. It allows more high-type borrowers to obtain unsecured loans; it may distort lending to low-type borrowers when the signal remains noisy; and it changes the bank's competitive response by altering the attractiveness of collateralized contracts. The baseline model therefore studies a conservative limiting case. It shuts down the inefficiency from lending to low types and shows that improved fintech information can reduce welfare even when data-based lending creates no misallocation across borrower types.

Creditworthy Low-Type Borrowers. The assumption that low-type borrowers hold non-creditworthy projects is not the driving force behind any of the welfare results.[50] When low-type borrowers instead have projects with positive NPV, they are served by an unsecured contract that earns both lenders zero profit. The rest of the equilibrium retains its baseline structure. In the Online Appendix, we show how the condition in equation (1) must be modified so that the bank can separate the two types through collateral in equilibrium. A change in fintech information then generates the same effects on payoffs and welfare as in the baseline model.

To conclude, our extensions establish that our results are not artifacts of stark assumptions but rather follow from an equilibrium substitution between data-based screening and collateral-based screening. Better information strengthens the competitive position

[50] This is also an important difference between our paper and previous papers that study open banking initiatives. In related papers, such as He et al. (2023, 2024), negative welfare effects arise because some borrowers are not creditworthy. Therefore, information may drive in or out socially costly borrowers.

of data-based lenders and weakens the position of collateral-specialist lenders. If collateralization is socially costly, competition can become destructive: borrowers receive more attractive private terms, while total surplus may fall because equilibrium lending relies too heavily on costly collateralization.

7 Conclusion

We study competition in a loan market with adverse selection between a bank and a fintech that are asymmetric along two key dimensions: their ability to screen borrowers and their capacity to extract value from collateralized assets in the case of default. Open banking and improvements in data analytics strengthen the fintech's competitive position, reduce adverse selection, and allow for the provision of more unsecured loans. However, they also intensify competition, compelling the bank to respond with more aggressive (i.e., higher collateral) offers. This trade-off lies at the heart of our analysis.

Our main contribution is to show that improvements in information are not always welfare-enhancing. While they reduce the inefficiencies associated with collateralization, they can also induce destructive competition and over-collateralization. As a result, the relationship between the informativeness of the fintech's signal and social welfare is non-monotonic. We establish conditions under which open banking enhances borrower surplus but reduces total welfare, and even cases where it is detrimental to fintechs themselves.

Despite the simplicity of our model, this trade-off persists in more general environments. Allowing the bank to also identify some borrowers would not overturn the result; it would lead the bank to offer unsecured loans to those it can identify, and the key distortion would still arise in the contested region where types are uncertain. Similarly, generalizing the signal structure to allow for false positives and false negatives introduces new sources of inefficiency, potentially strengthening our result. In particular, overly optimistic assessments by fintechs may lead to credit misallocation and higher defaults, reinforcing the non-monotonicity between information and welfare.

We also examine alternative institutional arrangements for data sharing, such as assigning property rights over data and establishing data markets. Our analysis shows that these alternatives can, in some cases, dominate open banking. For example, when the bank's comparative advantage in enforcing collateral is strong and the fintech's informational advantage is moderate, a regime where data is traded or controlled by lenders can outperform full data portability. These results resonate with recent discussions on markets for personal data and property rights over digital information.

From a policy perspective, our findings suggest that granting borrowers full control over their financial data does not guarantee welfare improvements. In environments where the costs of collateralization are high or legal institutions are weak, open banking may be beneficial. However, in well-developed financial systems with strong lender protections, unrestricted data sharing may actually reduce market efficiency. In such contexts, partial or regulated access to data, as well as institutional mechanisms that restrict the flow or usage of information, may improve outcomes. In this sense, our model calls for a more nuanced approach to data governance in credit markets–one that balances competition, information, and collateral constraints.

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Appendix

A Omitted Proofs

A.1 Notation

Payoffs and specific contracts.

- We denote the indifference curve of low-type borrowers where they have zero profit as $IC_0 = \{x \mid u^L(x) = 0\}$. This line partitions the contract space into three regions: the contracts on the line; the contracts on the right side of the line which are denoted by $IC_0^- = \{x \mid u^L(x) < 0\}$; and the contracts on the left side of the line denoted by $IC_0^+ = \{x \mid u^L(x) > 0\}$.

- Per-borrower welfare of a contract x between borrower i and lender ζ is denoted as $w^i(x)$. It is straightforward that, for contracts with no collateral $x = (R, 0)$, accepted by a high-type borrower, we have:

$$w^H(x) = \theta^H Y - 1 - r \equiv \bar{w}$$

The per-borrower profit of the lenders from such a contract with no collateral with a high-type borrower then writes $\pi_\zeta^H(x) = \bar{w} - u(x)$.

- For contracts $x_{IC_0} = (R, C)$ on the IC_0 line, we have $Y - R = \frac{1-\theta^L}{\theta^L}C$. The utility of the high-type borrower from such contract then writes:[51]

$$u(x_{IC_0}) = \theta^h \frac{1-\theta^L}{\theta^L}C - (1-\theta^H)C = C\frac{\theta^H - \theta^L}{\theta^L}.$$

The welfare of a contract on the IC_0 line between lender ζ and a borrower of high type then writes:

$$w(x_{IC_0}) = \bar{w} - (1-\theta^H)(1-\gamma_\zeta)C = \bar{w} - u(x_{IC_0})\frac{(1-\theta^H)(1-\gamma_\zeta)\theta^L}{\theta^H - \theta^L}.$$

We denote:

$$\beta^\zeta \equiv \frac{(1-\theta^H)(1-\gamma_\zeta)\theta^L}{\theta^H - \theta^L}.$$

We can then write $w(x_{IC_0}) = \bar{w} - \beta^\zeta u(x_{IC})$, and $\pi^i(x_{IC_0}) = \bar{w} - (1+\beta)u(x_{IC})$.

We can further simplify the notation by denoting $\beta = \beta^\zeta$.

- We define the function $G(.)$ such that it maps contracts on the IC_0 line to contracts with no collateral such that the high-type borrower is indifferent between them. Formally:

$$G : \mathbb{R}^2 \to \mathbb{R}^2 \text{ s.t if } x' = (R, 0) = G(x_{IC_0}), \text{ then } u(R, 0) = u(x_{IC_0}).$$

- We denote the contract on the IC_0 line for which the fintech earns zero profit when the high-type borrower accepts it by x_0^-. Formally x_0^- is such that $u^L(x_0^-) = 0$ and $\pi_F^H(x_0^-) = 0$. Then:

$$x_0^- = \left(\frac{1+r}{\theta^H}, \frac{\bar{w}\theta^L}{\theta^H(1-\theta^L)}\right).$$

We denote contract x_1^- on the IC_0 line such that $\pi_B^H(x_1^-) = (1-\sigma)\pi_B^H(x_0^-)$. Note that since $\gamma_B > \gamma_F = 0$, this contract exists and $\pi_B^H(x_1^-) > 0$ for $\sigma \neq 1$.

Strategies and Equilibrium. We denote lender ζ's strategy by $X_\zeta = (X_\zeta^+, X_\zeta^-)$, where X_ζ^+ (resp. X_ζ^-) is the loan contract offered by lender ζ to borrowers who receive a positive (resp. negative) signal from ζ. Contracts lie in a Borel set $M = M^+ \times M^-$ with Borel σ-algebra $\mathcal{B}(M)$.

[51] To simplify the notation, we use $u(\cdot)$ to denote the high-type borrower utility, and $u^L(\cdot)$ for low-type borrower utility.

A *pure strategy* of lender ζ is a pair $(x_\zeta^+, x_\zeta^-) \in M$. A *mixed strategy* of lender ζ is a Borel probability measure μ_ζ on M: for every $A \in \mathcal{B}(M)$,

$$\mathbb{P}\{(X_\zeta^+, X_\zeta^-) \in A\} = \mu_\zeta(A), \qquad \mu_\zeta(M) = 1.$$

A pure strategy is the special case $\mu_\zeta = \delta_{(x_\zeta^+, x_\zeta^-)}$. When μ_ζ is absolutely continuous it is represented by a joint density f_ζ with $\mu_\zeta(A) = \iint_A f_\zeta(x^+, x^-)\, dx^+ dx^-$; we do not restrict μ_ζ to this case. When lender ζ plays μ_ζ, the offered pair (X_ζ^+, X_ζ^-) is a random draw from μ_ζ: a borrower with a positive (negative) signal from ζ is offered X_ζ^+ (X_ζ^-).

We denote borrowers' strategy by $D^i = (d_B^i, d_F^i)$ such that $d_\zeta^i \in [0, 1]$ is the probability that a borrower of type i accepts the contract offered by lender ζ, and $d_B^i + d_F^i \in [0, 1]$.

We then denote the profit of lender ζ as $\Pi_\zeta(X_\zeta, X_{\zeta'}, D^i) = \Pi_\zeta^+(X_\zeta, X_{\zeta'}, D^i) + \Pi_\zeta^-(X_\zeta, X_{\zeta'}, D^i)$, where $\Pi_\zeta^+(X_\zeta, X_{\zeta'}, D^i)$ denotes the profit from offers to borrowers with a positive signal, and $\Pi_\zeta^-(X_\zeta, X_{\zeta'}, D^i)$ the profit from offers to borrowers with a negative signal, when the other lender plays $X_{\zeta'}$ and borrowers' choice of contract is D^i.

A PBE is denoted by $\hat{E} = (\hat{X}_B, \hat{X}_F, \hat{D}^H, \hat{D}^B)$ such that:

$$\hat{X}_\zeta = \operatorname{argmax}\{\Pi_\zeta(X_\zeta, X_{\zeta'}, D^i(X_\zeta, X_{\zeta'}))\},$$

$$\hat{D}^i(X_B, X_F) = \operatorname{argmax}\{d_B^i u^i(X_B^i) + d_F^i u^i(X_F^i)\}.$$

A.2 Proofs

Proof. of Proposition 1:

We characterize the unique class of equilibria in undominated strategies by a series of claims.

Claim 1. *In equilibrium, contracts accepted by borrowers must either have no collateral or lie on the IC_0 line.*

Proof of claim 1:

a) Consider an offer x_ζ^- to borrowers for whom lender ζ has a negative signal. First suppose $x_\zeta^- \in IC_0^-$. Since the IC_0 line is steeper than the iso-profit of the lenders for offers to high-type borrowers (ZP_ζ^H), there exists $x' \in IC_0$ with $u(x') = u(x_\zeta^-)$ and $\pi_\zeta^H(x') > \pi_\zeta^H(x_\zeta^-)$, a profitable deviation. If $x_\zeta^- \in IC_0^+$, all low-type borrowers accept x_ζ^- or the rival's negative-signal offer (whichever lies further left of IC_0). By assumption $\frac{r}{\theta^H} < Y < \frac{r}{\theta^\circ}$, at least one lender's profit on accepted contracts on IC_0^+ is negative, so at least one lender profitably deviates by offering no contract.

b) Consider an offer $x_\zeta^+ = (R, C)$ with $C > 0$ to borrowers with a positive signal. Since a positive signal ensures the borrower is of high type, $x' = (R, 0)$ satisfies $u(x') = u(x_\zeta^+)$ and $\pi_\zeta^H(x') > \pi_\zeta^H(x_\zeta^+)$: if x_ζ^+ is accepted with positive probability, offering x' is a profitable deviation.

Following claim 1, we restrict attention to equilibria in which every offered contract lies on the IC_0 line or on the zero-collateral axis: $M_\zeta^- \subseteq IC_0$ and $M_\zeta^+ \subseteq \{(R,0)\}$, where M_ζ^+ and M_ζ^- denote the supports of the marginals of μ_ζ, i.e. the sets of positive- and negative-signal contracts offered with positive probability. On both sets $u(\cdot)$ is injective, so we index contracts by the utility they deliver to high-type borrowers and represent the marginals by the CDFs

$$F_\zeta^+(t) = \mathbb{P}\{u(X_\zeta^+) \le t\}, \qquad F_\zeta^-(t) = \mathbb{P}\{u(X_\zeta^-) \le t\}, \qquad t \in \mathbb{R}^+,$$

which are non-decreasing and right-continuous, with mass points appearing as jumps. When no confusion arises we write $F_\zeta^s(x^s) := F_\zeta^s(u(x^s))$, $s \in \{+,-\}$, and $f_\zeta^s := \frac{d}{dt}F_\zeta^s$ where the derivative exists.

Claim 2. *There exists no pure strategy equilibrium.*

Proof of claim 2: Assume the fintech offers $X_F = (x_F^+, x_F^-)$ in pure strategy, with $u(x_F^+) > u(x_F^-)$. The bank's profit from any offer x_B writes:

$$\Pi_B(x_B, X_F) = \lambda \times \begin{cases} 0 & , \ u(x_B) < u(x_F^-) \\ \frac{1}{2}(1-\sigma)\pi_B^H(x_B) & , \ u(x_B) = u(x_F^-) \\ (1-\sigma)\pi_B^H(x_B) & , \ u(x_F^-) < u(x_B) < u(x_F^+) \\ (1-\frac{\sigma}{2})\pi_B^H(x_B) & , \ u(x_B) = u(x_F^+) \\ \pi_B^H(x_B) & , \ u(x_B) > u(x_F^+) \end{cases}$$

If $\pi_B^H(x_F^-) > 0$, $\Pi_B(\cdot, X_F)$ is strictly decreasing in $u(x_B)$ on each interval where the accepting pool is fixed and jumps down at the tie points. Its supremum, $\lambda \max\{(1-\sigma)\pi_B^H(x_F^-),\ \pi_B^H(x_F^+)\} > 0$, is approached as $u(x_B)$ descends to a tie point from above but never attained: the bank has no best reply. If instead $\pi_B^H(x_F^-) \le 0$, then $\gamma_B > \gamma_F$ implies $\pi_F^H(x_F^-) < \pi_B^H(x_F^-) \le 0$, so the fintech profitably withdraws x_F^- whenever it is accepted with positive probability. Preventing acceptance requires $u(x_B) \ge u(x_F^-)$: a tie still leaves x_F^- accepted with positive probability, while $u(x_B) > u(x_F^-)$ gives $\pi_B^H(x_B) < \pi_B^H(x_F^-) \le 0$ on every accepted borrower, and the bank profitably withdraws. The case $u(x_F^+) \le u(x_F^-)$ is analogous: the bank's profit is again discontinuous at the tie points and no x_B supports an equilibrium.

Claim 3. *Offering any contract x^- such that $\pi_\zeta^H(x^-) < 0$ is a weakly dominated strategy.*

Proof of claim 3: Such an offer yields negative profit if accepted by any borrower and zero otherwise, while any $x' \in IC_0$ with $\pi^H_\zeta(x') > 0$ yields positive or zero profit. Hence x' weakly dominates x^-.

Claim 4. *In a mixed strategy equilibrium,*

$$\text{Sup}\{u(x) \mid x \in M_F^-\} \;\leq\; \text{Inf}\{u(x) \mid x \in M_F^+\}.$$

Proof of claim 4: The fintech must be indifferent among all contracts in M_F^+:

$$\Pi_F^+(x^+, X_B) = \lambda\sigma F_B(x^+)\pi_F^H(x^+) = k, \;\; \forall x^+ \in M_F^+,$$

for some $k \geq 0$. First assume $k = 0$. Let x_{0B}^- satisfy $\pi_B^H(x_{0B}^-) = 0$. Then for $\gamma_B < 1$, there exists x^+ such that $u(x^+) > u(x_{0B}^-)$ and the fintech earns positive profit by offering with it, attracting all high-type borrowers.[52] Hence $k > 0$ and $F_B(x^+) = \frac{k}{\lambda\sigma(\bar{w}-u(x^+))}$ on M_F^+. For any x^- with $u(x^-) \in \{u(x) \mid x \in M_F^+\}$, the fintech's profit writes

$$\Pi_F^-(x^-, X_B) = \lambda(1-\sigma)F_B(x^-)\pi_F^H(x^-) = \frac{k(1-\sigma)(\bar{w} - (1+\beta^F)u(x^-))}{\sigma(\bar{w} - u(x^-))},$$

which is strictly decreasing in $u(x^-)$, since $\beta^F > 0$. Hence the fintech cannot be indifferent among such contracts and strictly prefers negative-signal offers with $u(x^-) \leq \text{Inf}\{u(x) \mid x \in M_F^+\}$.

Claim 5. *In a mixed strategy equilibrium in undominated strategies, the fintech earns zero profit from its offers to borrowers with a negative signal:* $\Pi_F^-(\hat{E}) = 0$.

Proof of claim 5: Suppose $\Pi_F^-(\hat{E}) > 0$. Every contract in M_F^- must earn the same strictly positive per-offer profit,

$$\lambda(1-\sigma)\, F_B^{\leftarrow}(x)\, \pi_F^H(x^-) = k_F > 0, \qquad \forall\, x \in M_F^-, \qquad (*)$$

where $F_B^{\leftarrow}(\cdot)$ denotes the left limit (a borrower strictly prefers the fintech). This implies that $\pi_F^H(x^-) > 0$, i.e. $u(x^-) < u(x_0^-)$, for every $x^- \in M_F^-$.

Let $\underline{u} \equiv \inf\{u(x) \mid x \in M_F^-\}$ and $A = \{x_B \mid u(x_B) \leq \underline{u}\}$. Letting $u(x^-) \to \underline{u}$ in $(*)$ gives $F_B(\underline{u}) > 0$, so $\mu_B(A) > 0$. Every contract in A yields zero profit to the bank: by claim 4, $\underline{u} \leq \sup\{u(x) \mid x \in M_F^-\} \leq \inf\{u(x) \mid x \in M_F^+\}$, so a contract with $u(x_B) < \underline{u}$ is accepted by no borrower, and one with $u(x_B) = \underline{u}$ could be accepted only through a tie with a fintech atom at $\underline{u}$. This case is precluded: no equilibrium has both lenders place an

[52]If such a contract attracts no high-type borrower, the bank must offer some x with $u(x) > u(x_{0B}^-)$ accepted by all of them, earning negative profit, and has a profitable deviation.

atom at the same utility, since the fintech would profitably shift its atom by ε, capturing all tied borrowers at a negligible loss in margin.

Now consider the deviation transferring this mass to x_0^-:

$$\mu'_B(S) = \mu_B(S \setminus A) + \mu_B(A)\, \delta_{x_0^-}(S) \qquad \text{for every measurable } S.$$

This deviation leaves the profit from offers outside A unchanged, while x_0^- (which is preferred to every negative-signal fintech offer since $u(x^-) < u(x_0^-)$ on M_F^-) is accepted by all negative-signal high types and earns $\lambda(1-\sigma)\,\pi_B^H(x_0^-) > 0$. This is a profitable deviation and contradicts $\Pi_F^-(\hat{E}) > 0$.

Claim 6. *In a mixed strategy equilibrium in undominated strategies,*

$$\Pi_B(\hat{E}) \geq \lambda(1-\sigma)\pi_B^H(x_0^-) = \lambda\pi_B^H(x_1^-).$$

Proof of claim 6: Suppose $\Pi_B(\hat{E}') = k < \lambda(1-\sigma)\pi_B^H(x_0^-)$ in some equilibrium $\hat{E}'$. Since the fintech plays no weakly dominated strategy, a bank offer $\dot{x}^-$ with $u(\dot{x}^-) > u(x_0^-)$ is accepted by all borrowers for whom the fintech has a negative signal. As $\dot{x}^- \to x_0^-$, the bank's profit approaches at least $\lambda(1-\sigma)\pi_B^H(x_0^-) > k$ which is a profitable deviation.

Claim 7. *In a mixed strategy equilibrium in undominated strategies,*

$$\Pi_F(\hat{E}) \geq \lambda\sigma\pi_F^H(G(x_1^-)).$$

Proof of claim 7: Suppose $\Pi_F^+(\hat{E}') = k < \lambda\sigma\pi_F^H(G(x_1^-))$ in some equilibrium $\hat{E}'$. Offering $G(x_1^-)$ yields

$$\Pi_F(G(x_1^-)) = \lambda\sigma F_B(x_1^-)\pi_F^H(G(x_1^-)) \leq k,$$

so $F_B(x_1^-) < 1$: the bank offers with positive probability some x^- with $u(x^-) > u(x_1^-)$, hence $\pi_B^H(x^-) < \pi_B^H(x_1^-)$. This contradicts claim 6, since such a contract earns less than $\lambda\pi_B^H(x_1^-)$ even if accepted by all high-type borrowers.

Claim 8. *The following mixed strategies constitute an equilibrium:*

$$M_B = \{x^- \mid u(x_0^-) \leq u(x) \leq u(x_1^-)\} \;\;,\;\; F_B(x^-) = \frac{\bar{w} - u(x_1^-)}{\bar{w} - u(x^-)},\;\; \forall x^- \in M_B,$$

$$M_F^+ = \{x^+ \mid u(x_0^-) \leq u(x) \leq u(x_1^-)\} \;\;,\;\; F_F^+(x^+) = \frac{(1-\sigma)(1+\beta)(u(x^+) - u(x_0^-))}{\sigma(\bar{w} - (1+\beta)u(x^+))},\;\; \forall x^+ \in M_F^+,$$

$$M_F^- = \{x^- \mid u(\underline{x}^-) \leq u(x) < u(x_0^-)\},\; \textit{for some } \underline{x}^- \textit{ s.t. } u(\underline{x}^-) < u(x_0^-),$$

and any proper CDF $F_F^-(\cdot)$ such that

$$F_F^-(x^-) \leq \frac{\bar{w} - (1+\beta)u(x_0^-)}{\bar{w} - (1+\beta)u(x^-)},\;\; \forall x^- \in M_F^-.$$

Proof of claim 8: We show there exists no profitable deviation.

a) **No deviation for the bank:** For any $x' \in M_B$, using $1-\sigma+\sigma F_F^+(x') = \frac{(1-\sigma)(\bar{w}-(1+\beta)u(x_0^-))}{\bar{w}-(1+\beta)u(x')}$,

$$\Pi_B(x', \hat{X}_F) = \lambda[1-\sigma+\sigma F_F^+(x')]\pi_B^H(x') = \lambda(1-\sigma)\pi_B^H(x_0^-) = \lambda\pi_B^H(x_1^-).$$

Hence the bank is indifferent on M_B. For x' on the IC_0 line with $u(x') < u(x_0^-)$,

$$\Pi_B(x', \hat{X}_F) = \lambda(1-\sigma)F_F^-(x')\pi_B^H(x') \leq \lambda(1-\sigma)\pi_B^H(x_0^-),$$

by the bound on $F_F^-(\cdot)$. For $u(x') > u(x_1^-)$, all high-type borrowers accept and $\Pi_B = \lambda\pi_B^H(x') < \lambda\pi_B^H(x_1^-)$. Hence no deviation is profitable.

b) **No deviation for the fintech:** Since $\mathrm{Sup}\{u(x) \mid x \in M_F^-\} \leq \min\{u(x) \mid x \in M_B\}$ and $F_F^-(\cdot)$ has no atom at x_0^-, the equilibrium offers X_F^- are accepted by no borrower and yield zero profit, in line with claim 5. A deviation to any x^- with $u(x^-) \geq u(x_0^-)$ yields $\pi_F^H(x^-) \leq 0$ and is not profitable. For any $x' \in M_F^+$,

$$\Pi_F(x', \hat{X}_B) = \lambda\sigma F_B(x')\pi_F^H(x') = \lambda\sigma(\bar{w} - u(x_1^-)) = \lambda\sigma\pi_F^H(G(x_1^-)).$$

Hence the fintech is indifferent on M_F^+. A contract x' with $u(x') < u(x_0^-)$ is accepted by no high-type borrower and yields zero profit, while $u(x') > u(x_1^-)$ yields $\Pi_F^+ = \lambda\sigma\pi_F^H(x') < \lambda\sigma\pi_F^H(G(x_1^-))$. No deviation is profitable.

Claim 9. *Any equilibrium in undominated mixed strategies has payoffs equivalent to the equilibrium characterized in claim 8.*

Proof of claim 9: Claims 5, 6, and 7 uniquely pin down the supports. If the bank offered x_B^- with $u(x_B^-) < u(x_0^-)$, the fintech could earn positive profit on negative-signal offers in $(u(x_B^-), u(x_0^-))$, violating claim 5; offering $u(x_B^-) > u(x_1^-)$ violates claim 6. Hence $M_B = \{x^- \mid u(x_0^-) \leq u(x) \leq u(x_1^-)\}$, and M_F^+ must coincide with it in utilities for both indifference conditions to hold. Given the supports, $F_B(\cdot)$ and $F_F^+(\cdot)$ are the unique distributions satisfying the indifference conditions. Only $F_F^-(\cdot)$ is not unique, but these offers are accepted by no borrower and serve only to deter bank deviations, so they do not affect payoffs.

□

Proof. of Proposition 2:

We first solve for payoffs and welfare in equilibrium, then analyze their behavior with respect to σ.

a) **Profits.** Since each lender is indifferent across its support, equilibrium profits equal their values at the extreme contracts:

$$\Pi_B(\hat{E}) = \lambda(1-\sigma)\pi_B^H(x_0^-) = \lambda(1-\sigma)[\bar{w} - (1+\beta)u(x_0^-)] \ ,$$

$$\Pi_F(\hat{E}) = \lambda\sigma\pi_F^H(G(x_1^-)) = \lambda\sigma[\bar{w} - u(x_1^-)].$$

Since $\pi_B^H(x_1^-) = \bar{w} - (1+\beta)u(x_1^-) = (1-\sigma)(\bar{w} - (1+\beta)u(x_0^-))$, we have $u(x_1^-) = u(x_0^-) + \sigma(\frac{\bar{w}}{1+\beta} - u(x_0^-))$. Denoting $u_0 \equiv u(x_0^-)$ and $u_1 \equiv u(x_1^-)$:

$$\Pi_B(\sigma) = \lambda(1-\sigma)[\bar{w} - (1+\beta)u_0] \ , \ \Pi_F(\sigma) = \lambda\sigma\left[\bar{w} - u_0 - \sigma\left(\frac{\bar{w}}{1+\beta} - u_0\right)\right].$$

b) **Welfare.** In equilibrium, all high-type borrowers accept a loan offer and low-type borrowers accept none. A contract x^+ with no collateral generates welfare $\bar{w}$; a contract x^- on the IC_0 line generates welfare $\bar{w} - \beta u(x^-)$, with inefficiency $\beta u(x^-)$. The social inefficiency of the equilibrium is therefore

$$SI(\hat{E}) \equiv \lambda\bar{w} - W(\hat{E}) = \int_{x_0^-}^{x_1^-} \lambda[1-\sigma+\sigma F_F^+(x^-)]\,\beta u(x^-)\,dF_B(x^-),$$

the inefficiency of the bank's IC_0 contracts accepted by high-type borrowers. From the proof of proposition 1, $F_B(\cdot)$ has a mass point $\mu^B(u_0) = \frac{\bar{w}-u_1}{\bar{w}-u_0}$ at u_0, and density $f^B(u) = \frac{\bar{w}-u_1}{(\bar{w}-u)^2}$ for $u \in (u_0, u_1]$, writing $u = u(x)$, and using

$$1 - \sigma + \sigma F_F^+(u) = \frac{(1-\sigma)(\bar{w} - (1+\beta)u_0)}{\bar{w} - (1+\beta)u},$$

we can write

$$SI(\hat{E}) = \lambda(1-\sigma)\beta u_0\frac{\bar{w}-u_1}{\bar{w}-u_0} + \lambda\beta(1-\sigma)(\bar{w}-u_1)(\bar{w}-(1+\beta)u_0)\int_{u_0}^{u_1}\frac{u\,du}{(\bar{w}-u)^2(\bar{w}-(1+\beta)u)}$$

$$= \lambda(1-\sigma)\beta u_0\frac{\bar{w}-u_1}{\bar{w}-u_0}$$

$$+\lambda(1-\sigma)(\bar{w}-u_1)(\bar{w}-(1+\beta)u_0)\left[\frac{1}{\beta\bar{w}}\ln\left(\frac{\bar{w}-u_1}{(1-\sigma)(\bar{w}-u_0)}\right) + \frac{1}{\bar{w}-u_0} - \frac{1}{\bar{w}-u_1}\right]$$

$$= \lambda(1-\sigma)((1+\beta)u_0 - u_1) + \frac{\lambda}{\beta\bar{w}}(1-\sigma)(\bar{w}-u_1)(\bar{w}-(1+\beta)u_0)\ln\left(\frac{\bar{w}-u_1}{(1-\sigma)(\bar{w}-u_0)}\right).$$

We denote $\alpha \equiv \frac{\bar{w}}{1+\beta} - u_0 > 0$, so that $u_1 = u_0 + \alpha\sigma$, and $A \equiv \bar{w} - u_0$, so that $\bar{w} - (1+\beta)u_0 = \alpha(1+\beta)$ and $\bar{w} - u_1 = A - \alpha\sigma$. Note that β is strictly decreasing and α strictly increasing in γ. Then:

$$W(\sigma) = \lambda\bar{w} - SI(\sigma) = \lambda\left[\bar{w} - (1-\sigma)\left(\beta u_0 - \alpha\sigma + \frac{\alpha(1+\beta)}{\beta\bar{w}}(A-\alpha\sigma)\ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right)\right)\right].$$

c) **Borrowers.** Substituting $\Pi_B(\sigma)$, $\Pi_F(\sigma)$ and $W(\sigma)$ into $U(\sigma) = W(\sigma) - \Pi_B(\sigma) - \Pi_F(\sigma)$ and simplifying yields

$$U(\sigma) = \lambda\left[u_0 + \alpha\sigma - \frac{\alpha(1+\beta)}{\beta\bar{w}}(A-\alpha\sigma)(1-\sigma)\ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right)\right].$$

1. Note that $\frac{A-\alpha\sigma}{A(1-\sigma)} > 1$: it is continuous and increasing in σ, going from 1 at $\sigma = 0$ to $+\infty$ at $\sigma = 1$. Hence $U(\sigma)$ is continuous. Differentiating and using $\frac{\alpha(1+\beta)}{\beta\bar{w}}(A-\alpha) = \alpha$:

$$U'(\sigma) = \frac{\alpha(1+\beta)}{\beta\bar{w}}(A+\alpha-2\alpha\sigma)\ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right).$$

Since $A+\alpha-2\alpha\sigma = (\bar{w}-u_1)+\alpha(1-\sigma) > 0$ and the logarithm is non-negative,

$$U'(\sigma) \geq 0 \ \ \forall\sigma \in [0,1].$$

2. Bank's profit, $\Pi_B(\sigma) = \lambda\alpha(1+\beta)(1-\sigma)$, is continuous and linearly decreasing in σ. Fintech's profit, $\Pi_F(\sigma) = \lambda\sigma(A-\alpha\sigma)$, is quadratic in σ with vertex at $\frac{A}{2\alpha}$, and has a local maximum in $(0,1)$ if and only if $\frac{A}{2\alpha} < 1$. As $\gamma \to 0$, $\alpha \to 0$ and $\frac{A}{2\alpha} \to +\infty$. When $\gamma \to 1$, $\beta \to 0$, $\alpha \to A$ and $\frac{A}{2\alpha} \to \frac{1}{2}$. Since $\frac{A}{2\alpha}$ is strictly decreasing in γ, there exists a threshold γ_F such that fintech's profit is strictly increasing for $\gamma < \gamma_F$ and has a local maximum in $(\frac{1}{2},1)$ for $\gamma > \gamma_F$.

3. First note that $W(1) = \bar{w} > W(0) = \bar{w} - \beta u_0$. The first derivative w.r.t. σ writes:

$$W'(\sigma) = \lambda\Bigg[\underbrace{(\beta u_0 - 2\alpha\sigma)}_{\frac{1}{\lambda}(\Pi_B'(\sigma)+\Pi_F'(\sigma))} + \underbrace{\frac{\alpha(1+\beta)}{\beta\bar{w}}(A+\alpha-2\alpha\sigma)\ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right)}_{\frac{1}{\lambda}U'(\sigma)}\Bigg].$$

We have $W'(0) = \lambda\beta u_0 > 0$ and $\lim_{\sigma\to1} W'(\sigma) = +\infty$. As shown above, $U'(\sigma) > 0$ for all σ, but $\Pi_B'(\sigma) + \Pi_F'(\sigma) = \beta u_0 - 2\alpha\sigma$ can become negative if α is sufficiently large (γ sufficiently large). The second derivative writes:

$$W''(\sigma) = -2\lambda\alpha\Bigg[1 + \underbrace{\frac{\alpha(1+\beta)}{\beta\bar{w}}\ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right)}_{Q_1(\sigma)} + \underbrace{\frac{(2\alpha\sigma-\alpha-A)}{2(A-\alpha\sigma)(1-\sigma)}}_{Q_2(\sigma)}\Bigg].$$

We show that $W''(\sigma)$ has a unique root:

$$Q_1'(\sigma) + Q_2'(\sigma) = -\frac{(A-\alpha)^2}{2(A-\alpha\sigma)^2(1-\sigma)^2} < 0, \quad \forall\sigma \in (0,1),$$

so $W''(\sigma)$ is strictly increasing, with

$$W''(0) = -2\lambda\alpha[1+Q_2(0)] = -2\lambda\alpha\frac{A-\alpha}{2A} = \frac{-\lambda\alpha\beta\bar{w}}{A(1+\beta)} < 0,$$

using $A - \alpha = \frac{\beta \bar{w}}{1+\beta}$. Also we have $\lim_{\sigma \to 1} W^{"}(\sigma) = +\infty$. Therefore $W^{"}(\sigma)$ has a unique interior root. We denote the root by $\sigma^* \in (0,1)$: $W(\sigma)$ is concave for $\sigma < \sigma^*$ and convex for $\sigma > \sigma^*$. Hence, if $W'(\sigma^*) > 0$, the welfare function is strictly increasing; if $W'(\sigma^*) < 0$, it has a local maximum at some $\sigma < \sigma^*$ and a local minimum at some $\sigma > \sigma^*$.

It remains to show that there exists a unique $\gamma_W \in (0,1)$ such that $W(\sigma)$ is strictly increasing if $\gamma \leq \gamma_W$ and bimodal if $\gamma > \gamma_W$. Define $\varepsilon \equiv A - \alpha = \frac{\beta \bar{w}}{1+\beta} > 0$. Recall that $\beta(\alpha + u_0) = \varepsilon$, so that $\beta u_0 = \frac{\varepsilon\, u_0}{\alpha + u_0}$, and that α is strictly increasing in γ, with $\frac{d\alpha}{d\gamma} = \frac{\beta^F \bar{w}}{(1+\beta)^2} > 0$, $\lim_{\gamma \to 0} \alpha = 0$ and $\lim_{\gamma \to 1} \alpha = A$. Define the normalized minimum slope:

$$m(\gamma) \equiv \frac{W'(\sigma^*(\gamma);\gamma)}{\lambda\, \varepsilon}.$$

Since $W^{"} < 0$ for $\sigma < \sigma^*$ and $W^{"} > 0$ for $\sigma > \sigma^*$, σ^* is the unique global minimizer of W' on $(0,1)$:

$$W'(\sigma) \;\geq\; \lambda \varepsilon\, m(\gamma) \;\; \forall \sigma \in (0,1), \;\; \text{with equality only at } \sigma = \sigma^*.$$

Hence, if $m(\gamma) \geq 0$, then $W' \geq 0$ with at most one zero and W is strictly increasing; if $m(\gamma) < 0$, then $W'(\sigma^*) < 0$ and W is bimodal. We prove three claims: (i) $\lim_{\gamma \to 0} m(\gamma) = 1$; (ii) $m(\gamma) < 0$ for γ sufficiently close to 1; (iii) $m(\gamma)$ is continuously differentiable and strictly decreasing on $(0,1)$.

Step 1. Since $U'(\sigma) \geq 0$, we have $W'(\sigma)/\lambda \geq \beta u_0 - 2\alpha\sigma \geq \beta u_0 - 2\alpha$, while $W'(0)/\lambda = \beta u_0$. Dividing by ε and

using $\frac{\beta u_0}{\varepsilon} = \frac{u_0}{\alpha + u_0}$:

$$\frac{u_0}{\alpha + u_0} - \frac{2\alpha}{\varepsilon} \;\leq\; m(\gamma) \;\leq\; \frac{u_0}{\alpha + u_0}.$$

As $\gamma \to 0$, $\alpha \to 0$ and $\varepsilon \to A$, so both bounds converge to 1. Hence $\lim_{\gamma \to 0} m(\gamma) = 1$.

Step 2. Fix any $\tilde{\sigma} \in (0,1)$, independent of γ. Since σ^* minimizes W',

$$m(\gamma) \;\leq\; \frac{W'(\tilde{\sigma};\gamma)}{\lambda \varepsilon}.$$

We expand this upper bound, compute its limit as $\gamma \to 1$ holding $\tilde{\sigma}$ fixed, and then choose $\tilde{\sigma}$ such that the limit is negative. Write

$$\frac{W'(\tilde{\sigma})}{\lambda} = (\beta u_0 - 2\alpha\tilde{\sigma}) + H\, L\, \ln\left(\frac{A - \alpha\tilde{\sigma}}{A(1-\tilde{\sigma})}\right) \;,\;\; H \equiv \frac{\alpha(1+\beta)}{\beta \bar{w}} = \frac{\alpha}{\varepsilon} \;,\;\; L \equiv A + \alpha - 2\alpha\tilde{\sigma},$$

and define

$$z \equiv \frac{\varepsilon\, \tilde{\sigma}}{A(1-\tilde{\sigma})} \;,\;\; \text{so that } \frac{A - \alpha\tilde{\sigma}}{A(1-\tilde{\sigma})} = 1 + z.$$

Note that as $\gamma \to 1$, $z \to 0$ at any fixed $\tilde{\sigma}$. For the first term in the upper bound, $\frac{\beta u_0}{\varepsilon} = \frac{u_0}{\alpha+u_0}$ gives

$$\frac{\beta u_0 - 2\alpha\tilde{\sigma}}{\varepsilon} = \frac{u_0}{\alpha + u_0} - \frac{2\alpha\tilde{\sigma}}{\varepsilon}.$$

For the second term in the upper bound, we use $\ln(1+z) = z - \frac{z^2}{2} + O(z^3)$. For the term linear in z, substituting $L = 2A(1-\tilde{\sigma}) - \varepsilon(1-2\tilde{\sigma})$ yields:

$$\frac{H\,L\,z}{\varepsilon} = \frac{\alpha\,L\,\tilde{\sigma}}{\varepsilon\,A(1-\tilde{\sigma})} = \frac{2\alpha\tilde{\sigma}}{\varepsilon} - \frac{\alpha\tilde{\sigma}(1-2\tilde{\sigma})}{A(1-\tilde{\sigma})}.$$

For the term quadratic in z, using $L = 2A(1-\tilde{\sigma}) + O(\varepsilon)$ yields:

$$\frac{H\,L\,z^2}{2\varepsilon} = \frac{\alpha\,L\,\tilde{\sigma}^2}{2A^2(1-\tilde{\sigma})^2} = \frac{\alpha\tilde{\sigma}^2}{A(1-\tilde{\sigma})} + O(\varepsilon).$$

The remainder of the second term in the upper bound is $\frac{H\,L}{\varepsilon}\,O(z^3) = \frac{\alpha L}{\varepsilon^2}\,O(\varepsilon^3) = O(\varepsilon)$. Collecting the pieces and simplifying, we get:

$$\frac{W'(\tilde{\sigma})}{\lambda\varepsilon} = \frac{u_0}{\alpha+u_0} - \frac{\alpha\tilde{\sigma}}{A} + O(\varepsilon) \quad\Longrightarrow\quad \lim_{\gamma\to 1}\frac{W'(\tilde{\sigma};\gamma)}{\lambda\varepsilon} = \frac{u_0}{\bar{w}} - \tilde{\sigma}.$$

This limit is negative if and only if $\tilde{\sigma} > \frac{u_0}{\bar{w}}$, and since $\frac{u_0}{\bar{w}} = \frac{1}{1+\beta^F} < 1$, such $\tilde{\sigma}$ exists. Choosing $\tilde{\sigma} = \frac{1}{2}(1+\frac{u_0}{\bar{w}})$, the limit equals $-\frac{A}{2\bar{w}} < 0$. Hence there exists $\bar{\gamma} < 1$ such that

$$m(\gamma) \;\le\; \frac{W'(\tilde{\sigma};\gamma)}{\lambda\varepsilon} \;<\; -\frac{A}{4\bar{w}} \;<\; 0 \;\; \forall\gamma \in (\bar{\gamma}, 1).$$

Hence $W(\sigma)$ is bimodal for all γ sufficiently close to 1.

Step 3. We show that $m(\gamma)$ is continuously differentiable and strictly decreasing on $(0,1)$. This formalizes the idea that as γ rises, once W becomes bimodal it cannot revert to strictly monotone.

Since $Q_1'(\sigma) + Q_2'(\sigma) < 0$ strictly, σ^* is a simple root of W'', and by the implicit function theorem $\sigma^*(\gamma)$ is continuously differentiable. Hence $m(\gamma)$ is continuously differentiable and, since $W''(\sigma^*) = 0$, only the direct dependence on α survives ($\bar{w}$, u_0, and A are independent of γ):

$$m'(\gamma) = \frac{d\alpha}{d\gamma}\cdot\frac{E(\sigma^*)}{\varepsilon^2} \;\;,\;\; E(\sigma) \equiv \varepsilon\,\frac{\partial}{\partial\alpha}\Big(\frac{W'(\sigma)}{\lambda}\Big) + \frac{W'(\sigma)}{\lambda},$$

Since $\frac{d\alpha}{d\gamma} > 0$, it suffices to show $E(\sigma^*) < 0$. Using $W''(\sigma^*) = 0$, we can write:

$$H\ln\Big(\frac{A-\alpha\sigma^*}{A(1-\sigma^*)}\Big) = \frac{L(\sigma^*)}{2P(\sigma^*)} - 1 \;\;,\;\; L(\sigma) \equiv A+\alpha-2\alpha\sigma \;\;,\;\; P(\sigma) \equiv (A-\alpha\sigma)(1-\sigma). \qquad (\star)$$

Using $(\star)$ we can write[53]

$$E(\sigma^*) = \frac{\varepsilon\, R(\sigma^*, \alpha, A)}{2\alpha\,(A - \alpha\sigma^*)(1 - \sigma^*)} - \frac{u_0\, \varepsilon^2}{(\alpha + u_0)^2}\,,$$

where

$$R(\sigma, \alpha, A) \equiv 4\alpha^2\sigma^3 - 6\alpha^2\sigma^2 + 2\alpha^2\sigma - 6A\alpha\sigma^2 + 8A\alpha\sigma - 3A\alpha + 2A^2\sigma - A^2.$$

The second term of $E(\sigma^*)$ is strictly negative for every $u_0 > 0$. Hence the proof boils down to showing that $R(\sigma^*, \alpha, A) < 0$ at σ^*. Establishing this inequality is mostly tedious algebra and we therefore relegate it to the Online Appendix (Lemma 2). Granting this Lemma, $E(\sigma^*) < 0$ and $m'(\gamma) < 0$ for all $\gamma \in (0, 1)$.

By Steps 1–3, m is continuous and strictly decreasing on $(0, 1)$, positive near 0 and negative near 1. Hence there exists a unique $\gamma_W \in (0, 1)$ with $m > 0$ on $(0, \gamma_W)$, $m(\gamma_W) = 0$ and $m < 0$ on $(\gamma_W, 1)$. For $\gamma \leq \gamma_W$, $W' \geq 0$ with at most one zero, so $W(\sigma)$ is strictly increasing; for $\gamma > \gamma_W$, $W'(\sigma^*) < 0$, so $W(\sigma)$ has exactly one interior local maximum on $(0, \sigma^*)$ and one interior local minimum on $(\sigma^*, 1)$. This concludes part 3 of the proposition.

□

Proof. of Proposition 3:

1. It follows directly from proposition 2.

2. It follows directly from proposition 2.

3. From proposition 2, $\Pi_F(\sigma)$ is strictly increasing for $\gamma < \gamma_F$: then $\hat{\sigma}^F = \bar{\sigma}$ and the fintech prefers open banking with full data sharing. For $\gamma > \gamma_F$, Π_F is inverse U-shaped with maximum at $\sigma^F_{max} \in (0, 1)$, and three cases are possible:

   a) If $\sigma^F_{max} \geq \bar{\sigma}$, then $\hat{\sigma}^F = \bar{\sigma}$: the fintech prefers open banking with full data sharing.

   b) If $\sigma^F_{max} \leq \underline{\sigma}$, then $\hat{\sigma}^F = \underline{\sigma}$: the fintech prefers closed banking.

   c) If $\sigma^F_{max} \in (\underline{\sigma}, \bar{\sigma})$, the fintech prefers partial data sharing with $\sigma = \sigma^F_{max}$.

[53]The simplification differentiates $\frac{W'(\sigma)}{\lambda} = \beta u_0 - 2\alpha\sigma + H\, L(\sigma) \ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right)$ with respect to α, substituting $\beta = \frac{A-\alpha}{\alpha+u_0}$ and $\frac{\partial}{\partial\alpha} \ln\left(\frac{A-\alpha\sigma}{A(1-\sigma)}\right) = \frac{-\sigma}{A-\alpha\sigma}$, then eliminates the logarithm using $(\star)$.

4. From proposition 2, $W(\sigma)$ is strictly increasing for $\gamma < \gamma_W$: then $\hat{\sigma}_W = \bar{\sigma}$ and open banking with full data sharing maximizes welfare. For $\gamma > \gamma_W$, W is bimodal with local maximum at σ_W^{max} and local minimum at σ_W^{min}, decreasing on $(\sigma_W^{max}, \sigma_W^{min})$, and three cases are possible:

   a) If $\bar{\sigma} \leq \sigma_W^{max}$ or $\underline{\sigma} \geq \sigma_W^{min}$, then $\hat{\sigma}_W = \bar{\sigma}$: open banking with full data sharing maximizes welfare.

   b) If $\sigma_W^{max} \in (\underline{\sigma}, \bar{\sigma})$, open banking with full data sharing maximizes welfare only if $W(\bar{\sigma}) \geq W(\sigma_W^{max})$; otherwise partial data sharing with $\sigma = \sigma_W^{max}$ is socially optimal.

   c) If $\sigma_W^{max} < \underline{\sigma}$, open banking with full data sharing maximizes welfare only if $W(\bar{\sigma}) \geq W(\underline{\sigma})$; otherwise closed banking is socially optimal.

□

Proof. of Lemma 1:
Recall from the proof of Proposition 2 that:

$$\Pi_B(\sigma) = \lambda\alpha(1+\beta)(1-\sigma),$$

$$\Pi_F(\sigma) = \lambda\sigma(A - \alpha\sigma).$$

Therefore,

$$\Pi'_B(\sigma) + \Pi'_F(\sigma) = -\alpha(1+\beta) + A - 2\alpha\sigma = \beta u_0 - 2\alpha\sigma,$$

where the simplification is done by replacing $A - \alpha = \frac{\beta\bar{w}}{1+\beta}$. Since $\beta u_0 - 2\alpha\sigma$ is strictly decreasing in γ (positive at $\gamma \to 0$ and negative at $\gamma \to 1$), there exists a threshold γ_{BF} (a threshold in α) at which: $\Pi'_B(1) + \Pi'_F(1) = 0$.

Hence, for $\gamma \leq \gamma_{BF}$ the joint profit is strictly increasing, and for $\gamma > \gamma_{BF}$ the joint profit is inverse-U shaped and has a local maximum at some $\sigma \in (0, 1)$.

Recall that $\Pi_B(\sigma)$ is always strictly decreasing, while $\Pi_F(\sigma)$ is strictly increasing for $\gamma \leq \gamma_F$ and inverse-U shaped for $\gamma > \gamma_F$. This implies that, for $\gamma = \gamma_{BF}$, we have $\Pi'_F(\sigma) > 0$ for all σ. Therefore we must have $\gamma_{BF} < \gamma_F$.

Similarly, since $U(\sigma)$ is strictly increasing for $\gamma < \gamma_W$, for $\gamma = \gamma_{BF}$, we have $W'(\sigma) > 0$ for all σ. Therefore we must have $\gamma_{BF} < \gamma_W$.

□

Proof. of Proposition 4:

1. If $\gamma < \gamma_{BF} < \gamma_W$, welfare and joint profit are both strictly increasing. Hence, $\hat{\sigma}_{BF} = \hat{\sigma}_W = \bar{\sigma}$, and the same outcome is obtained under any allocation of property rights.

2. If $\gamma \in (\gamma_{BF}, \gamma_W)$, welfare is strictly increasing in σ, while the joint profit has a local maximum at some $\sigma_{BF}^{max} \in (0,1)$. If $\sigma_{BF}^{max} \in (\underline{\sigma}, \bar{\sigma})$, then $\hat{\sigma}_{BF} = \sigma_{BF}^{max} < \hat{\sigma}_W = \bar{\sigma}$, and assigning property rights to one of the lenders result in lower welfare than open banking with full data sharing.

3. If $\gamma > \gamma_W > \gamma_{BF}$, welfare is bimodal and the joint profit is inverse-U shaped in σ. If $\hat{\sigma}_{BF} = \hat{\sigma}_W = \bar{\sigma}$, the allocation of property rights is irrelevant. Otherwise, open banking is socially optimal if $W(\hat{\sigma}_W) > W(\hat{\sigma}_{BF})$, and allocating property rights to one of the lenders is socially optimal if $W(\hat{\sigma}_W) < W(\hat{\sigma}_{BF})$.

□

Proof. of Proposition 5:

1. If $\gamma < \min\{\gamma_F, \gamma_W\}$, both fintech's profit and welfare are strictly increasing in σ, and hence $\hat{\sigma}_F = \hat{\sigma}_W = \bar{\sigma}$. In this case, fintech chooses the welfare maximizing level of signal informativeness.

2. Assume $\gamma \in [\gamma_F, \gamma_W]$. In that case, fintech's profit is inverse U-shaped and welfare is strictly increasing in σ. If $\hat{\sigma}_F < \bar{\sigma} = \hat{\sigma}_W$, the fintech under-invests in data analytics.

3. Assume $\gamma \in [\gamma_W, \gamma_F]$. In this case, welfare is bimodal and fintech's profit is strictly increasing in σ. If $\hat{\sigma}_W < \bar{\sigma} = \hat{\sigma}_F$, the fintech over-invests in data analytics.

4. If $\gamma > \max\{\gamma_F, \gamma_W\}$, fintech's profit is inverse U-shaped and welfare is bimodal in σ. If $\hat{\sigma}_W = \hat{\sigma}_F$, fintech's incentives aligns with those of society. Otherwise, the fintech either under-invests (when $\hat{\sigma}_W > \hat{\sigma}_F$) or over-invests (when $\hat{\sigma}_W < \hat{\sigma}_F$).

□

B Online Appendix

This appendix examines the robustness of our results by relaxing the simplifying assumptions of the baseline model and extending the analysis to richer environments.

B.1 Bank's Data Analytics

In this section, we consider an environment in which the bank can also use borrowers' data to assess their creditworthiness. We assume that both lenders share an identical signal structure that produces no false positives. A lender's signal quality is given by $\sigma_\zeta = \sigma_\zeta^H(S^+)$, the probability that a high-type borrower receives a good signal from lender ζ. For low-type borrowers, we set $\sigma_\zeta^L(S^+) = 0$, which implies that both lenders can perfectly identify a low-type borrower. Moreover, we assume that lenders' signals are stochastically independent.[54]

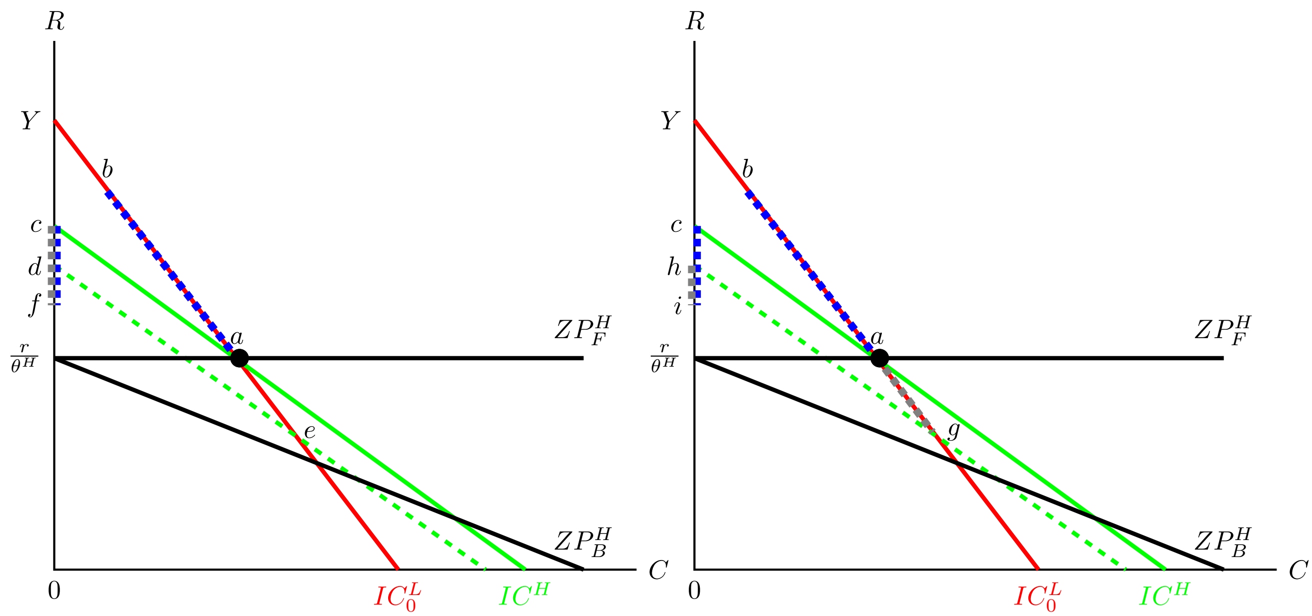
Equilibrium

Let us begin with the case in which the fintech's signal is fully uninformative ($\sigma_F = 0$), while the bank can identify some high-type borrowers ($\sigma_B > 0$). Similar to the baseline equilibrium, the bank can prevent the fintech from earning positive profits from attracting high-type borrowers with contract a at the intersection of the low-type's zero indifference line (IC_L) and the fintech's zero-profit line ZP_F^H, as shown in Figure 6a.

However, because the bank can identify some borrowers as high-type, it can earn higher profits by offering them a non-collateralized loan. Thus, the bank offers contract c to borrowers who receive a positive signal from the bank. This contract provides high-type borrowers with the same utility as contract a, which prevents the fintech from attracting them. In equilibrium, the fintech offers randomized collateralized contracts in $(a, b]$, which removes any profitable deviations for the bank.

Now assume that the fintech can also identify some high-type borrowers ($\sigma_F > 0$). The fintech can attract these borrowers by offering a non-collateralized contract below c. The bank can respond by offering contracts below a and c to compete with the fintech. However, because the bank has some captive borrowers—those who receive a negative signal at the fintech—it will offer better terms only as long as its profit does not fall below what it earns from offering contracts a and c to these captive borrowers.

[54] The independence of signals in competition among lenders is a widely used assumption in the literature, as in Broecker (1990), Hauswald and Marquez (2006), and He et al. (2023), reflecting the idea that lenders' creditworthiness tests use proprietary data and information-processing technologies. Nonetheless, the same qualitative results can be obtained with correlated signals, as long as the correlation is not perfect.



(a) $\sigma_F < \sigma_B$

(b) $\sigma_F \geq \sigma_B$

Figure 6: **Graphical representation of the equilibrium.**

The key to understanding the equilibria of this game is that, for any precision level of the fintech's signal σ_F, the bank is willing to offer better terms to borrowers for whom it receives a positive signal. Consider contract e on the IC_0 line such that, if accepted by all borrowers with a negative signal at the bank ($\lambda(1-\sigma_B)$), it yields the same profit as contract a when accepted only by borrowers who receive a negative signal from both lenders ($\lambda(1-\sigma_B)(1-\sigma_F)$). Similarly, consider the non-collateralized contract f such that, if accepted by all borrowers with a positive signal at the bank ($\lambda\sigma_B$), it generates the same profit as contract c when accepted only by borrowers who receive a positive signal at the bank and a negative signal from the fintech ($\lambda\sigma_B(1-\sigma_F)$). As illustrated in Figure 6a, contract f provides higher utility to high-type borrowers than e. This implies that, if the fintech offers f, the bank is indifferent between matching this contract for borrowers with a positive signal and offering c, but is strictly better off not matching it for borrowers with a negative signal.

Now suppose that the bank's offer to borrowers with a negative signal at the bank is fixed at a, so the fintech can attract those with a negative signal at the bank and a positive signal at the fintech ($\lambda\sigma_F(1-\sigma_B)$) by offering a contract slightly below c. In this case, if the bank does not offer better collateralized loans, these borrowers become captives of the fintech. Thus, if the bank offers non-collateralized loans below c, the fintech is willing to match them only if the resulting profit is not lower than the profit it can earn by capturing borrowers with a negative signal at the bank and a positive signal at the fintech using a contract slightly below c.

Let us denote by x the contract that, if accepted by all borrowers with a positive signal at the fintech ($\lambda\sigma_F$), generates the same profit as contract c when it is accepted by borrowers with a bad signal at the bank and a good signal at the fintech ($\lambda\sigma_F(1-\sigma_B)$). Recall that contract f is defined such that, if accepted by all borrowers with a positive signal at the bank ($\lambda\sigma_B$), it yields the same profit as contract c when accepted only by borrowers who receive a positive signal at the bank and a negative signal from the fintech ($\lambda\sigma_B(1-\sigma_F)$). It is straightforward to verify that, for $\sigma_F < \sigma_B$, contract x lies below contract f, implying that the fintech is more aggressive in competing for borrowers with a good signal. In contrast, when $\sigma_F > \sigma_B$, contract x lies above f, implying that the fintech is not willing to match f if the bank's offer to borrowers with a negative signal is fixed at a. This observation is central to understanding the discontinuity in equilibria when σ_F rises above σ_B.

As illustrated in Figure 6a, for $\sigma_F < \sigma_B$, the fintech offers randomized contracts on $[c, f]$ such that the bank is indifferent to offering any non-collateralized loan to borrowers with a positive signal within this interval. As noted above, in this case the bank is strictly better off not matching these offers with collateralized loans and instead offering a to borrowers with a negative signal.

In contrast, for $\sigma_F \geq \sigma_B$, the fintech is not willing to offer contract f when the bank's non-collateralized offer is fixed at a. In this case, as illustrated in Figure 6b, the bank offers randomized collateralized loans on the interval $[a, g]$ and randomized non-collateralized loans on the interval $[h, i]$, such that the fintech is indifferent to offering any contract on $[c, i]$ to borrowers with a positive signal. Similarly, the fintech's randomized offer makes the bank indifferent between offering collateralized loans on $[a, g]$ and non-collateralized loans on $[h, i]$.

The following proposition summarizes these equilibria in undominated strategies. The proof of Proposition 6 in the appendix fully characterizes the supports and the distribution of randomized loan offers.

Proposition 6. *There exists a mixed-strategy equilibrium in undominated strategy where borrowers with a positive signal receive a strictly better offer than borrowers with a negative signal, and:*

1. *For $\sigma_F < \sigma_B$, the fintech offers loans backed up by collateral to borrowers with the negative signal and non-collateralized loans to borrowers with the positive signal, and the bank offers one collateralized loan to borrowers with negative signal and non-collateralized loans to*

borrowers with the positive signal.

2. *For $\sigma_F \geq \sigma_B$, both lenders offer randomized loans backed up by collateral to borrowers with the negative signal and non-collateralized loans to borrowers with the positive signal.*

Welfare Effects of Information

In this section, we examine the effects of improved information on equilibrium outcomes and welfare. In this setting, improvements in information can correspond to an increase in the fintech's signal precision σ_F, an increase in the bank's signal quality σ_B, or both. We first analyze how a higher σ_F affects equilibrium outcomes and then consider the impact of an increase in σ_B. The effect of simultaneous increases in both parameters can be interpreted as the superposition of these two effects.

We begin by examining the effect of an increase in σ_F on borrowers' utility $U(\sigma)$. As in the baseline analysis, higher signal precision enables the fintech to identify and attract a larger share of high-type borrowers, which triggers a competitive response from the bank. The resulting increase in competition intensity unambiguously raises the utility of high-type borrowers; high-type borrowers strictly prefer a higher signal precision for the fintech.

Next, consider the effect of an increase in σ_F on lenders' profits. As in the baseline analysis, the bank's profit is strictly decreasing in the fintech's signal quality, since a higher σ_F allows the fintech to attract a larger share of borrowers who were previously captive to the bank. The fintech's profit, however, may be non-monotonic in its signal precision. Although a higher σ_F increases the share of borrowers the fintech can attract, the per-borrower profit from lending to these borrowers declines with σ_F, because the bank responds with more competitive offers.

This negative effect of σ_F on the fintech's profit, operating through intensified competition, is in fact stronger when the bank possesses data-analytics capability. Recall from the baseline analysis that, when $\sigma_B = 0$, the fintech's profit becomes non-monotonic in σ_F when γ is sufficiently large—that is, when the bank can respond to improvements in the fintech's data analytics by offering collateralized loans that are more attractive to high-type borrowers. When $\sigma_B > 0$, the bank can match any offer the fintech makes to borrowers with a positive signal. Thus, a higher $\sigma_B > 0$ implies a more competitive non-collateralized response from the bank to improvements in the fintech's signal quality.

For $\sigma_F < \sigma_B$, since the bank's collateralized loan offer is fixed at a, fintech's profit can have an inverse U-shape if σ_B is sufficiently high, regardless of γ. As σ_F rises above σ_B, the competitive response of the bank depends on its ability to match fintech's offers with collateralized loans, and hence fintech's profit for $\sigma_F > \sigma_B$ becomes decreasing in σ_F only if bank's ability to respond with collateralized loans is sufficiently high, i.e., γ is large enough[55].

Given that bank's profit is strictly decreasing in fintech's signal precision, and fintech's profit might be non-monotonic in σ_F, the joint profit of the lenders can also have an inverse U-shape. This is particularly more likely when bank's signal precision σ_B, or its ability in capturing collateral value γ, is sufficiently high, resulting in more intense competition when the informativeness of the fintech's signal improves.

Now consider the effect of σ_F on total welfare. Recall from the baseline analysis that an increase in the fintech's signal precision can reduce welfare by intensifying collateralization in bank loans. As shown in Figure 6a, this mechanism is absent when $\sigma_F < \sigma_B$, since the bank's offer to borrowers with a negative signal is fixed at a. In these equilibria, total welfare increases unambiguously with σ_F, implying that the rise in borrowers' utility outweighs the decline in the bank's profit and any potential decline in the fintech's profit.

However, when σ_F rises above σ_B, the bank responds to the fintech's improved information by offering collateralized loans with better terms for high-type borrowers, that is, loans with lower interest rates and higher collateral requirements. As in the baseline analysis, this intensified collateralization in the bank's offers to borrowers with a bad signal creates a trade-off for total welfare: while it increases welfare by expanding the use of non-collateralized loans, it also reduces welfare by increasing the collateral cost in bank loans. This trade-off can generate non-monotonic welfare as a function of σ_F, particularly when the bank's competitive response is strong, that is, when γ is sufficiently high.

Proposition 7. *The following results hold regarding the effect of σ_F on payoffs and welfare:*

1. ***(Borrowers)*** *The payoff of the borrowers, $U(\sigma_F)$, is continuous and strictly increasing in σ_F.*

2. ***(Lenders)*** *The bank's profit, $\Pi_B(\sigma_F)$, is continuous and strictly decreasing in σ_F. Moreover,*

    (a) For $\sigma_F < \sigma_B$, the fintech's profit $\Pi_F(\sigma_F)$ is continuous and strictly increasing in σ_F if $\sigma_F \leq \frac{1}{2}$, and is inverse-U shaped in σ_F if $\sigma_F > \frac{1}{2}$.

[55] Note that the discontinuity in the equilibria at $\sigma_F = \sigma_B$ implies that lenders' and borrowers' payoffs may exhibit a kink at this level of signal precision. As a result, the fintech's profit as a function of σ_F may have two local maxima, particularly when both σ_B and γ are sufficiently high: one in $[0, \sigma_B]$ and another in $[\sigma_B, 1]$.

(b) For $\sigma_F \geq \sigma_B$, there exists a threshold $\gamma_F \in (0,1)$ such that: for $\gamma \leq \gamma_F$, the fintech's profit $\Pi_F(\sigma_F)$ is strictly increasing in σ_F, and for $\gamma > \gamma_F$, the fintech's profit is inverse-U shaped in σ_F.

3. ***(Joint Profit)*** *There exist threshold $\gamma^f_{BF_1}$ and $\gamma^f_{BF_2}$ such that:*

   (a) For $\sigma_F \leq \sigma_B$, the sum of lenders profit $\Pi_B(\sigma_F) + \Pi_F(\sigma_F)$ is strictly increasing in σ_F if $\gamma \leq \gamma^f_{BF_1}$ and has an inverse-U shape if $\gamma > \gamma^f_{BF_1}$.

   (b) For $\sigma_F > \sigma_B$, the sum of lenders profit $\Pi_B(\sigma_F) + \Pi_F(\sigma_F)$ is strictly increasing in σ_F if $\gamma \leq \gamma^f_{BF_2}$ and has an inverse-U shape if $\gamma > \gamma^f_{BF_2}$.

4. ***(Total Welfare)***

   (a) Total welfare $W(\sigma_F)$ is continuous, and strictly increasing in $\sigma_F \in [\sigma_F, \sigma_B]$.

   (b) For $\sigma_F \geq \sigma_B$, there exists a threshold $\gamma_W \in (0,1)$ such that: for $\gamma \leq \gamma_W$, total welfare $W(\sigma_F)$ is strictly increasing in σ, and for $\gamma > \gamma_W$, total welfare $W(\sigma_F)$ is bimodal, exhibiting an interior local minimum and an interior local maximum in $\sigma_F \in (\sigma_B, 1)$.

We now examine the effect of an improvement in the bank's data analytics, that is, an increase in σ_B, on equilibrium outcomes.

First, consider borrowers' payoffs and suppose that σ_B increases within the interval $[0, \sigma_F]$, for some given $\sigma_F > 0$. As the bank's ability to identify high-type borrowers through data processing improves, it relies less on collateralized loans. As shown in Figures 6a and 6b, the bank's loans offered to borrowers with a positive signal (in $[h, i]$) provide strictly higher utility to high-type borrowers than the loans offered to borrowers with a bad signal (in $[a, g]$). In response, the fintech offers loans in $[h, i]$ with higher probability to compete for borrowers who receive a positive signal from both lenders. This implies that borrowers with a positive signal from at least one lender obtain better loan contracts. At the same time, borrowers with a negative signal from both lenders experience lower payoffs, as the bank's collateralized offers become less competitive when point g moves toward a as σ_B increases in $[0, \sigma_F]$.

Overall, the effect of an increase in σ_B in the interval $[0, \sigma_F]$ on borrowers' ex-ante payoffs is positive: the increase in utility for borrowers with at least one positive signal outweighs the decline in payoffs for borrowers who receive negative signals from both lenders.

However, the increase in borrowers' payoffs ceases once σ_B reaches σ_F. As shown in Figure 6a, for $\sigma_B > \sigma_F$, the bank's offer to borrowers with a bad signal is fixed at a, and the support of lenders' offers to borrowers with at least one positive signal is determined by

σ_F.[56] In this case, an increase in σ_B enlarges the mass point at c in the bank's randomized offers to borrowers with a positive signal. Because contracts c and a yield identical payoffs for high-type borrowers, and the fintech does not respond with more competitive loan offers, borrowers' payoffs remain constant for $\sigma_B \in [\sigma_F, 1]$.

The key observation from the above analysis is that the fintech's response to an increase in the bank's signal precision depends on the relative strength of the fintech's signal quality. This observation helps clarify how lenders' profits are affected by σ_B. When $\sigma_F \geq \sigma_B$, an increase in σ_B leads the bank to offer better contracts to borrowers who receive a positive signal, which in turn triggers a competitive response from the fintech. In contrast, when $\sigma_F < \sigma_B$, an increase in σ_B induces the bank to offer c instead of a to borrowers with a positive signal. This change does not affect the probability that these borrowers accept the fintech's offers in $[c, f]$. As a result, while the fintech's profit declines as the bank's signal quality increases over $\sigma_B \in [0, \sigma_F]$, it remains constant for $\sigma_B \in [\sigma_F, 1]$.

We now consider the effect of an increase in σ_B on the bank's profit. For $\sigma_B \in [0, \sigma_F]$, the bank's profit may be non-monotonic. The underlying reason is that, as σ_B increases over this interval, the bank offers better non-collateralized loans to a larger share of borrowers who receive a positive signal at the bank. The fintech responds with more competitive offers, which reduces the bank's per-borrower profit from lending to borrowers with a positive signal. This effect is stronger when replacing collateralized loans with non-collateralized ones does not generate high marginal profits for the bank, which occurs when the bank's cost of collateral is low. Consequently, for $\sigma_B \in [0, \sigma_F]$, the bank's profit can exhibit an inverse-U shape when γ is sufficiently high.

In contrast, for $\sigma_B \in [\sigma_F, 1]$, an improvement in the bank's signal precision does not lead to contracts with higher payoffs for borrowers and does not trigger a competitive response from the fintech. In this case, the bank instead replaces some offers at a with offers at c. Eliminating the collateral requirement for these loans increases the bank's profit. Therefore, over this interval, the bank's profit is strictly increasing in σ_B.

Finally, consider the effect of σ_B on total welfare. Recall that, in equilibrium, no low-type borrower accepts a loan, and thus the only source of inefficiency in the model is the welfare loss associated with collateralized loans. As discussed above, an increase in the bank's signal precision leads to less collateralization by the bank, regardless of the fintech's signal precision. This implies that total welfare is increasing in σ_B over the entire domain. The following proposition summarizes the effect of the bank's signal precision

[56] Recall that f is defined such that the bank is indifferent between offering c to borrowers who receive a good signal from the bank and a bad signal from the fintech ($\lambda\sigma_B(1-\sigma_F)$) and offering f to all borrowers with a positive signal at the bank.

on equilibrium outcomes.

Proposition 8. *The following results hold regarding the effect of σ_B on payoffs and welfare:*

1. ***(Borrowers)*** *The payoff of the borrowers, $U(\sigma_B)$, is continuous and strictly increasing in $\sigma_B \in [0, \sigma_F]$, and is constant for $\sigma_B \in [\sigma_F, 1]$.*

2. ***(Lenders)*** *The fintech's profit, $\Pi_F(\sigma_B)$, is strictly decreasing in $\sigma_B \in [0, \sigma_F]$, and is constant for $\sigma_B \in [\sigma_F, 1]$. Moreover,*

   (a) *For $\sigma_B \in [0, \sigma_F]$, there exists a threshold $\gamma_B \in (0, 1)$ such that: for $\gamma \leq \gamma_B$, the bank's profit $\Pi_B(\sigma_B)$ is strictly increasing in σ_B, and for $\gamma > \gamma_B$, the bank's profit is inverse-U shaped in σ_B.*

   (b) *For $\sigma_B \in [\sigma_F, 1]$, the bank's profit $\Pi_B(\sigma_B)$ is strictly increasing in σ_B.*

3. ***(Joint Profit)***

   (a) *For $\sigma_B \in [0, \sigma_F]$, there exist a threshold γ^b_{BF} such that the sum of lenders profit $\Pi_B(\sigma_B) + \Pi_F(\sigma_B)$ is strictly increasing in σ_B if $\gamma \leq \gamma^b_{BF}$, and has an inverse-U shape if $\gamma > \gamma^b_{BF}$.*

   (b) *For $\sigma_B \in [\sigma_F, 1]$, the lenders' joint profit $\Pi_B(\sigma_B) + \Pi_F(\sigma_B)$ is strictly increasing in σ_B.*

4. ***(Total Welfare)*** *Total welfare $W(\sigma_B)$ is strictly increasing in σ_B.*

Implications

In Section 5, we discussed several implications of our results for the effects of improvements in the fintech's data analytics on market outcomes. The results established in this section show that these implications also hold in a more general setting in which the bank possesses data-analytics capabilities.

Because the effects of σ_F on payoffs when $\sigma_B > 0$ are similar to those in the baseline model, three main conclusions can be generalized. First, full data sharing in an open banking regime that allocates data property rights to borrowers may not always be welfare enhancing. Particularly, when borrowers' data sharing raises the fintech's signal precision above that of the bank ($\bar{\sigma}_F > \sigma_B$) and collateral is highly valuable to the bank (γ is sufficiently high), partial data-sharing can be optimal. Second, in such environments, allocating data property rights to lenders through an established data market can outperform an open banking regime. Third, the fintech's incentives to invest in data-analytics improvements do not necessarily align with social welfare; the fintech may either over-invest or under-invest in data analytics, especially when the bank's advantage in collateralization is sufficiently strong.

Let's consider the implications of regulatory or technological innovations that enhance the bank's data-analytics capabilities. The key insight from this section is that improvements in the bank's data analytics—particularly through increased data flows to the bank—are always welfare enhancing. The difference between the welfare implications of data availability for the bank and for the fintech lies in the role of collateralization. While additional data for the fintech can reduce welfare by increasing collateral requirements through intensifying competition, better information for the bank always lowers its reliance on collateral and therefore unambiguously improves welfare.

Assume that the flow of data is from the fintech to the bank, that is, the fintech has access to the data that the bank can not possess in a closed banking regime. Since borrowers' payoffs are non-decreasing in the bank's signal precision, an open banking regime with full data sharing is always optimal in this setting.[57] Moreover, when access to data improves the bank's data analytics, allocating data property rights to lenders is weakly outperformed by allocating property rights to borrowers.

Finally, our results show that the bank may under-invest in data analytics. Although welfare is strictly increasing in σ_B, the bank's profit can be non-monotonic in its signal precision. Proposition 8 implies that under-investment is more likely when the bank is highly efficient in collateralization. In contrast to the fintech, the bank cannot over-invest in data analytics, since improvements in the bank's information always enhance welfare, even when collateral is highly valuable.

B.2 Multiple Banks and Fintechs

Our baseline model focuses on competition between a single collateral-dependent bank and a single data-driven fintech. In this section, we extend the model to several banks and fintechs and examine the robustness of our main results.

Multiple Banks

A critical assumption of our model is the bank's comparative advantage in using collateral. This advantage is the source of the bank's market power, and the destructive-competition channel operates through it. When the fintech obtains more data, the bank

[57] Note that, although borrowers' payoffs may be constant in σ_B, the fintech's profit is also constant over the same range of σ_B. The bank can therefore induce full data sharing by offering borrowers an arbitrarily small reward, without triggering a competitive response from the fintech.

exploits its collateral advantage by raising collateral requirements, which can reduce welfare. Perfect competition among symmetric banks with identical collateral costs shuts down this channel. Symmetric banks earn no rent on collateral, so there is no margin to dissipate through more aggressive offers. However, the mechanism survives under imperfect competition among banks that differ in their ability to extract value from heterogeneous borrowers' assets. As long as one bank retains a collateral advantage over its rivals, improvements in the fintech's information can still trigger destructive competition and welfare loss.

Consider a spatial model of bank competition in which banks are located at unit distances from one another along a line, and borrowers are distributed in this space. In the baseline model, a bank's per-unit collateral value γ is a single parameter. We now let it vary across borrowers, reflecting the idea that a bank extracts more value from the assets of borrowers it knows well, through local expertise or relationship lending.[58] We capture this through location: for a borrower between two adjacent banks, we let t_i and t_j denote her distance from each bank, with $t_i + t_j = 1$. A borrower at distance t_i from bank i delivers a per-unit collateral value $\gamma_i = \gamma(t_i)$, where $\gamma(\cdot)$ is decreasing. A borrower located closer to a bank thus yields that bank a higher collateral value. For simplicity, we assume that the fintech holds no such advantage and obtains $\gamma_F = 0$ from any borrower. The fintech has the same signal structure as in the baseline model, while the banks have no access to creditworthiness signals.

Consider first the case $\sigma = 0$. At any location t, the closer bank, say bank i, holds the collateral advantage $\gamma_i > \gamma_j$. In equilibrium, bank i offers the collateralized contract that leaves bank j with zero profit. The distant bank thus plays the same role as the fintech in the baseline model, acting as a competitive threat that pins down the closer bank's offer. Now suppose $\sigma > 0$. The fintech identifies a share of high-type borrowers and competes for them directly. Competition between the closer bank and the fintech then mirrors the baseline model. The closer bank randomizes over collateralized contracts, while the fintech randomizes over unsecured contracts.

The main results of the baseline model extend to this setting. Following the baseline analysis, the shape of welfare in σ depends on the collateral value of the winning bank. In markets sufficiently close to the winning bank i, where γ_i is high, welfare can be non-monotone in σ. In markets sufficiently far from the winning bank, where γ_i is low, welfare

[58] Our formulation is consistent with empirical evidence that proximity lowers the costs of assessing, inspecting, and repossessing collateral, as documented by Bellucci et al. (2019). The relevant distance need not be physical. Paravisini et al. (2023) show that banks specialize in particular lending markets, which confers a lending advantage and market power. Distance can thus be interpreted more broadly as proximity in a space of expertise, with a bank more proficient in the segments closest to it.

is monotone in σ.[59]

The overall effect of the fintech's information on welfare then depends on the range of $\gamma(\cdot)$ and the distribution of borrowers across these markets. In particular, open banking with full data sharing can be suboptimal relative to partial data sharing, or to allocating property rights to lenders and establishing a data market. This case arises when γ is sufficiently high as distance approaches zero and the share of borrowers in the banks' vicinity is sufficiently high. Open banking can thus be outperformed by alternative arrangements when proficient banks closely cover most market segments. By contrast, when banks do not closely cover all segments, so that many borrowers face a high collateral cost with every bank, open banking with full data sharing outperforms the alternative arrangements.

Multiple Fintechs

In this section, we extend the model to competition between a traditional bank and multiple fintechs. We retain the baseline assumption that the bank has no informative signal about borrowers' type. We consider N identical fintechs with the same characteristics as in the baseline model. That is, we assume that fintechs capture no value from collateral, $\gamma_F = 0$, and that they receive a partially informative signal about borrowers' type, which allows false negatives but not false positives, $\sigma^L_\zeta(S^+) = 0$ and $\sigma^H_\zeta(S^+) = \sigma \in [0, 1]$. Moreover, we assume that the fintechs' signals are independent.[60] We characterize the equilibrium of this game in the following proposition.

Proposition 9. *Assume $\sigma \in (0, 1)$. There exists a mixed-strategy equilibrium in undominated strategies in which the bank offers randomized loans backed by collateral, whereas the fintechs offer randomized loans backed by collateral to borrowers with the negative signal and non-collateralized loans to borrowers with the positive signal. In equilibrium, only the high-type borrowers borrow either from the bank with collateral or from the fintech without collateral. All lenders earn positive profit for $\sigma \in (0, 1)$.*

The equilibrium is qualitatively similar to the baseline. When $\sigma = 0$, the bank offers contract a, at the intersection of the IC_0 line and the fintechs' zero-profit line (Figure 2b), and the fintechs randomize over contracts on the IC_0 line above a, deterring any bank deviation. As σ rises, the fintechs attract high-type borrowers with a positive signal through

[59]The baseline results on the shape of the welfare and profit functions do not depend on the fintech's value of collateral, which in this extension is replaced by the collateral value of the adjacent bank. The proofs of Propositions 1 and 2 show that these results hold as long as $\gamma_F < \gamma_B$. The normalization $\gamma_F = 0$ is only a simplifying assumption.

[60]The same qualitative results can be obtained with correlated signals, as long as the correlation is not perfect.

unsecured contracts, and the bank can respond with separating collateralized contracts on the IC_0 line that offer these borrowers better terms.

However, the bank can still guarantee profit $(1-\sigma)^N \pi_B(a)$ by offering a, accepted by borrowers without a positive signal from any fintech. Let e satisfy $\pi_B(e) = (1-\sigma)^N \pi_B(a)$; the bank never offers contracts with per-borrower profit below $\pi_B(e)$. The equilibrium is again in mixed strategies: the bank randomizes over collateralized contracts in $[a, e]$ on the IC_0 line, while the fintechs randomize over unsecured contracts that deliver high types the same utility as the bank's contracts in $[a, e]$.

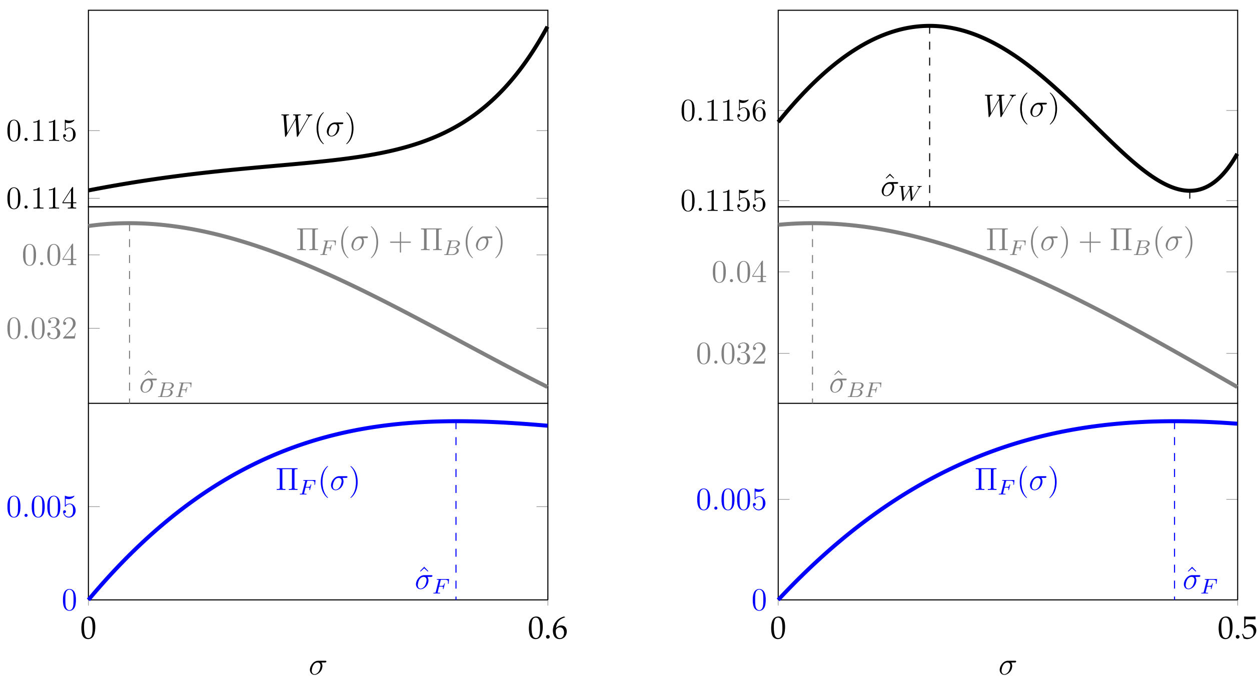
The main result of our baseline model, namely the possible adverse effect of better fintech information on welfare and its implications for open banking, data property rights, and investment in data analytics, rests on three features of the equilibrium. First, while borrowers' utility is increasing in σ, welfare can be non-monotone and decrease over intermediate values of σ when the bank's proficiency in using collateral, γ, is sufficiently high. Second, the lenders' joint profit can be non-monotone in σ, and the signal quality that maximizes it can yield higher or lower welfare than full data sharing, which is the level that maximizes borrowers' utility. Third, the fintech's profit can be non-monotone in σ, so that the fintech's incentive to invest in data analytics need not align with that of society. Our numerical simulations show that all three features carry over to the extended model with N fintechs.[61]

Figure 7 depicts two numerical examples for the competition between a bank and two fintechs, showing welfare, joint profit, and fintech profit as functions of the fintech's signal quality. In panel (a), welfare is increasing in σ, while joint profit and fintech profit attain interior maxima over the signal feasibility interval $[\underline{\sigma}, \overline{\sigma}]$. Here, open banking with full data sharing yields higher welfare than partial data sharing or than allocating data property rights to lenders in a data market. Moreover, the fintechs under-invest in data analytics, since their profit peaks at a signal quality below the highest attainable level $\overline{\sigma}$.

In panel (b), the bank is more proficient in using collateral (γ is higher) and welfare becomes non-monotone in σ. Here, open banking with partial data sharing, or allocating data property rights to lenders in a data market, outperforms open banking with full data sharing. In this case, the fintechs over-invest in data analytics, since their profit peaks at a signal quality that yields lower welfare than the level that maximizes welfare on $[\underline{\sigma}, \overline{\sigma}]$.

Remark 4. Taken together, the two extensions show that the main findings of our baseline model are robust to a richer market structure with multiple competing banks and

[61] While we report numerical simulations only for the case with two fintechs, we observe the same qualitative results in simulations with a larger number of fintechs.



(a) $\gamma = 0.88$

(b) $\gamma = 0.92$

Figure 7: **Competition between two fintechs and a bank; Welfare, joint profit, and fintech profit as functions of σ**

(a) $Y = 2$, $\theta^H = 0.62$, $\theta^L = 0.4$, $\lambda = 0.6$, $[\underline{\sigma}, \bar{\sigma}] = [0, 0.5]$, $\gamma = 0.88$. (b) $Y = 2$, $\theta^H = 0.62$, $\theta^L = 0.4$, $\lambda = 0.5$, $[\underline{\sigma}, \bar{\sigma}] = [0, 0.5]$, $\gamma = 0.92$

fintechs. The mechanism survives as long as competition remains imperfect on both sides of the market. Two ingredients sustain this imperfection. First, the bank's market power in credit markets with collateral persists in a spatial model with many banks, since proximity preserves a collateral advantage over rivals. Second, the fintechs' creditworthiness signals are not perfectly correlated, so each fintech retains an informational rent and all lenders earn positive profits as long as no signal is perfectly informative. The destructive-competition channel, and with it the non-monotonicity of welfare in σ, therefore does not depend on the market being a duopoly.

B.3 General Signal Structure

Our model focuses on a false-negative signal structure. While this assumption simplifies the analysis, it neither drives nor strengthens our results. To see this, consider the baseline model in which the bank does not possess data-analytics capability. Suppose that the fintech's signal structure allows for both false negatives and false positives.[62] In this case, if the fintech offers a non-collateralized loan to borrowers who receive a good signal, some low-type borrowers will also be granted loans. This introduces an additional source of inefficiency into the model: reliance on data analytics by the fintech can generate loan contracts for low-type borrowers who would otherwise be perfectly screened out through collateral requirements.

In the baseline model, where the fintech's signal does not generate false positives, the fintech can attract borrowers and earn positive profits for any $\sigma_F^H > 0$. This property no longer holds when false positives are allowed. For any $\sigma_F^L(S^+)$, the fintech can attract borrowers by offering a non-collateralized loan only if $\sigma_F^H(S^+)$ exceeds a threshold such that the average repayment probability of borrowers who receive a good signal is sufficiently high.

In this case, an improvement in the fintech's data analytics that raises $\sigma_F^H(S^+)$ above this threshold has three effects on welfare. First, it increases welfare by enabling non-collateralized loans to some high-type borrowers. Second, it reduces welfare by extending non-collateralized loans to some low-type borrowers. Third, it lowers welfare through the bank's competitive response, as the bank offers loans with lower interest rates and higher collateral requirements. These observations suggest that allowing false positives in the fintech's signal structure can strengthen our results regarding the potential negative welfare effects of improved information for the fintech.

[62]The fintech's signal quality can be represented by $(\sigma_F^H(S^+), \sigma_F^L(S^+))$, where $\sigma_F^H(S^+)$ is the probability that a high-type borrower receives a good signal from the fintech, and $\sigma_F^L(S^+)$ is the probability that a low-type borrower receives a good signal, with $\sigma_F^L(S^+) < \sigma_F^H(S^+)$.

Alternatively, improvements in the fintech's information can be interpreted as a reduction in $\sigma_F^L(S^+)$. While lowering the probability of false positives reduces the inefficiency associated with granting loans to low-type borrowers, it can still reduce welfare by intensifying collateral requirements in the bank's more competitive loan offers. Hence, better information for the fintech—whether through a reduction in false positives or false negatives—can generate greater inefficiencies arising from bank collateralization. In this sense, our model focuses on a limiting case in which the inefficiency from lending to low-type borrowers is normalized to zero and shows that, even in the absence of such inefficiency, improved information for the fintech can reduce welfare solely through intensified collateral requirements in bank loans.

B.4 The NPV of Low-type borrowers

In the baseline model, we assume that low-type borrowers possess projects with negative net present value. This assumption is a normalization and does not affect our results. Consider instead the case in which the NPV of low-type projects is positive, that is:

$$\frac{r}{\theta^H} < \frac{r}{\theta^L} < Y.$$

First, suppose that the fintech does not have access to an informative signal about borrowers' types, that is, $\sigma_F = 0$. Let x denote the non-collateralized contract that yields zero profit to the lender when accepted by low-type borrowers, $x = (\frac{r}{\theta^L}, 0)$. Next, consider the set of contracts that render low-type borrowers indifferent between those contracts and contract x.

$$\theta^L(Y - R) - (1 - \theta^L)C + \bar{C} = Y\theta^L - r + \bar{C} \tag{IC^L}$$

Let y denote the contract at the intersection of this line and the fintech's zero-profit line. By offering contracts x and y, the bank can perfectly separate the two types while eliminating any opportunity for the fintech to earn positive profits from either type, assuming that the share of high-type borrowers in the prior is sufficiently low.[63]

[63]Denote by y' the non-collateralized contract that yields the same utility to the high-type borrower as y. Then the sufficient condition for the separating equilibrium to sustain is that offering y' to all borrowers yields negative profit to the lenders. Formally, y and y' write:

$$y = (\frac{r}{\theta^H}, \frac{r + \theta^L(\frac{r}{\theta^H} - Y)}{1 - \theta^L}),$$

$$y' = (R', 0), \quad \text{where} \quad R' = \frac{r}{\theta^H} + \frac{1 - \theta^H}{\theta^H}(\frac{r + \theta^L(\frac{r}{\theta^H} - Y)}{1 - \theta^L}).$$

Then λ must be sufficiently low such that

$$\frac{r}{\theta^\circ} > R'.$$

Consequently, one can construct an equilibrium in undominated strategies in which both lenders offer x, the bank additionally offers y, and the fintech offers randomized contracts on the IC^L line above y to eliminate any profitable deviation by the bank. For any $\sigma_F > 0$, the same equilibrium structure as in the baseline model can be sustained by adding contract x to the set of bank offers and the contracts offered by the fintech to borrowers with a negative signal.

In these equilibria, for any $\sigma_F \in [0, 1]$, low-type borrowers accept contract x, while the bank's collateralized offers to high-type borrowers and the fintech's non-collateralized offers to borrowers with a positive signal depend on σ_F. A change in σ_F therefore generates the same effects on payoffs and welfare as in the baseline model.

B.5 Proofs

We use the following lemma in the proof of Proposition 2.

Lemma 2. *At any inflection point $\sigma^* \in (0,1)$ of the welfare function, $R(\sigma^*, \alpha, A) < 0$, where*

$$R(\sigma, \alpha, A) \equiv 4\alpha^2\sigma^3 - 6\alpha^2\sigma^2 + 2\alpha^2\sigma - 6A\alpha\sigma^2 + 8A\alpha\sigma - 3A\alpha + 2A^2\sigma - A^2.$$

Proof. of Lemma 2:
Write $s \equiv \sigma^*$ and let $z \equiv \frac{\varepsilon s}{A(1-s)} > 0$, so that $1 + z = \frac{A-\alpha s}{A(1-s)}$ and the inflection identity $(\star)$ reads $H \ln(1+z) = \frac{L(s)}{2P(s)} - 1$. The proof proceeds in three parts.

Part 1: $\sigma^* > \frac{1}{2}$.

For all $z > 0$, $\ln(1+z) > \frac{z}{1+z}$. Together with the algebraic identity

$$\frac{L(s)}{2P(s)} - 1 - H\,\frac{z}{1+z} = \frac{\varepsilon\,(2s-1)}{2P(s)},$$

$(\star)$ implies $\frac{\varepsilon(2s-1)}{2P(s)} > 0$, i.e., $s > \frac{1}{2}$.

Part 2: $J(s, \alpha, A) < 0$ at the inflection point, where

$$J(\sigma, \alpha, A) \equiv \alpha^2\sigma^3 + 2A\alpha\sigma^3 - 7A\alpha\sigma^2 + 2A\alpha\sigma - 2A^2\sigma^2 + 7A^2\sigma - 3A^2.$$

For all $z > 0$, the $[2/1]$ Padé approximation of $\ln(1+z)$ at $z = 0$ provides the upper bound $\ln(1+z) < \frac{z(z+6)}{4z+6}$.[64] Together with the algebraic identity $(\star)$

$$H\,\frac{z(z+6)}{4z+6} - \Big(\frac{L(s)}{2P(s)} - 1\Big) = \frac{-\,\varepsilon\,J(s,\alpha,A)}{2A(1-s)(A-\alpha s)(A(3-s)-2\alpha s)},$$

in which $(1-s) > 0$, $(A - \alpha s) > 0$ and $A(3-s) - 2\alpha s > A(3-s) - 2As = 3A(1-s) > 0$. Hence $(\star)$ implies that $J(s, \alpha, A) < 0$.

Part 3: for $s \in (\frac{1}{2}, 1)$, $J(s, \alpha, A) < 0$ implies $R(s, \alpha, A) < 0$.

Fix $s \in (\frac{1}{2}, 1)$ and view R and J as quadratic polynomials in α on $[0, A]$. R is concave in α (leading coefficient $2s(2s-1)(s-1) < 0$) with $R(s, 0, A) = A^2(2s-1) > 0$ and $R(s, A, A) = 4A^2(s-1)^3 < 0$; hence there is a unique $\alpha_R(s) \in (0, A)$, continuous in s, such that $R < 0$ if and only if $\alpha > \alpha_R(s)$. J is convex in α (leading coefficient $s^3 > 0$) with $J(s, 0, A) = A^2(2s-1)(3-s) > 0$ and $J(s, A, A) = 3A^2(s-1)^3 < 0$; hence there is a unique $\alpha_J(s) \in (0, A)$ such that $J < 0$ if and only if $\alpha > \alpha_J(s)$. The resultant of R and J with respect to α is

$$\mathrm{Res}_\alpha(R, J) = A^4\, s^3(s-1)^3(2s-1)(4s-5),$$

[64] The difference $\frac{z(z+6)}{4z+6} - \ln(1+z)$ vanishes at $z = 0$ and has derivative $\frac{4z^3}{(4z+6)^2(1+z)} > 0$.

which has no root in $(\frac{1}{2}, 1)$; hence R and J have no common zero with $s \in (\frac{1}{2}, 1)$. In particular, $s \mapsto J(s, \alpha_R(s), A)$ is continuous and never vanishes on $(\frac{1}{2}, 1)$, so it has constant sign. As $s \to \frac{1}{2}$, $\alpha_R(s) \to 0$ (since $R(\frac{1}{2}, \alpha, A) = -\frac{A\alpha}{2}$ vanishes only at $\alpha = 0$), and a first-order expansion gives $\alpha_R(s) = 2A(2s-1) + O((2s-1)^2)$, hence

$$J(s, \alpha_R(s), A) = \frac{3}{2}A^2(2s-1) + O((2s-1)^2) > 0$$

for s close to $\frac{1}{2}$. Therefore $J(s, \alpha_R(s), A) > 0$ on all of $(\frac{1}{2}, 1)$, which implies $\alpha_R(s) < \alpha_J(s)$: since $J(s, \alpha, A) \leq 0$ for all $\alpha \geq \alpha_J(s)$, then $J(s, \alpha_R(s), A) > 0$ rules out $\alpha_R(s) \geq \alpha_J(s)$.

By Parts 1 and 2, the inflection point satisfies $s > \frac{1}{2}$ and $J(s, \alpha, A) < 0$; by Part 3, $\alpha > \alpha_J(s) > \alpha_R(s)$, hence $R(s, \alpha, A) < 0$. □

Proof. of Proposition 6:
We define contracts x_1^- and x_1^+ such that:

$$\pi_B^H(x_1^-) = \frac{1-\sigma_F}{1-\sigma_B}\pi_B^H(x_0^-), \text{ and}$$

$$\pi_\zeta^H(x_1^+) = (1-\sigma_B)\pi_\zeta^H(H(x_1^-)),$$

where, as in the proof of Proposition 1, x_0^- is such that $u^L(x_0^-) = 0$ and $\pi_F^H(x_0^-) = 0$. We also define contract x_2^+ such that:

$$\pi_\zeta^H(x_2^+) = (1-\sigma_F)\pi_\zeta^H(H(x_0^-)).$$

1. We show that, for $\sigma_F < \sigma_B$, the following strategies construct an equilibrium:

$$X_B^- = x_0^-,$$

$$M_B^+ = M_F^+ = \{x^+ \mid u(x_0^-) \leq u(x) \leq u(x_2^+)\},$$
$$F_B^+(x^+) = \frac{(1-\sigma_F)(\bar{w} - u(x_0^-))}{\sigma_B(\bar{w} - u(x^-))} - \frac{1-\sigma_B}{\sigma_B}, \ \forall x^+ \in M_B^+, \sigma_B \neq 0,$$
$$F_F^+(x^+) = \frac{(1-\sigma_F)(u(x^+) - u(x_0^-))}{\sigma_F(\bar{w} - u(x^+))}, \ \forall (x^+, x^-) \in M_F, \sigma_F \neq 0,$$
$$M_F^- = \{x^- \mid u(\underline{x}^-)) \leq u(x) < u(x_0^-)\}, \text{ for some } \underline{x}^- \text{ s.t } u(\underline{x}^-) < u(x_0^-),$$

and any proper CDF $F_F^-(\cdot)$ such that:

$$F_F^-(x^-) \leq \frac{\bar{w} - (1+\beta)u(x_0^-)}{\bar{w} - (1+\beta)u(x^-)}, \ \forall (x^+, x^-) \in M_F.$$

a) **No Deviation for the Bank:** The bank's profit from borrowers with a negative signal is

$$\Pi_B^-(x_0^-, \hat{X}_F) = \lambda(1-\sigma_B)(1-\sigma_F)\pi_B^H(x_0^-) = \lambda(1-\sigma_B)(1-\sigma_F)(\bar{w}-(1+\beta)u(x_0^-)).$$

Note that $F_F^-(.)$ is the same as in proposition 1 and eliminates the possibility of any profitable deviation for the bank by offering x^- such that $u(x^-) < u(x_0^-)$. If the bank offers any x'^- such that $u(x'^-) > u(x_0^-)$, its profit writes:

$$\Pi_B^-(x'^-, \hat{X}_F) = \lambda(1-\sigma_B)\Big(1-\sigma_F+\sigma_F F_F^+(x'^-)\Big)(\bar{w}-(1+\beta)u(x'^-))$$

$$= \lambda(1-\sigma_B)(1-\sigma_F)\frac{\bar{w}-u(x_0^-)}{\bar{w}-u(x'^-)}(\bar{w}-(1+\beta)u(x'^-)) < \lambda(1-\sigma_B)(1-\sigma_F)(\bar{w}-(1+\beta)u(x_0^-)),$$

where the last inequality follows from $\frac{\bar{w}-u(x_0^-)}{\bar{w}-u(x'^-)} < \frac{\bar{w}-(1+\beta)u(x_0^-)}{\bar{w}-(1+\beta)u(x'^-)}$, for $u(x'^-) > u(x_0^-)$, and $\beta > 0$. Therefore, the bank has no profitable deviation for its offer to borrowers with a negative signal.

Now consider the bank's offers to borrowers with a positive signal. For any $x^+ \in M_B^+$:

$$\Pi_B^+(x^+, \hat{X}_F) = \lambda\sigma_B(1-\sigma_F+\sigma_F F_F^+(x^+))\pi_B^H(x^+) = \lambda\sigma_B(1-\sigma_F)(\bar{w}-u(x_0^-)).$$

Hence, the bank is indifferent between offering any contract in M_B^+. If the bank offers any x'^+ such that $u(x'^+) < u(x_0^-)$, its profit writes:

$$\Pi_B^+(x'^+, \hat{X}_F) = \lambda\sigma_B(1-\sigma_F)F_F^-(u(x'^+))\pi_B^H(x'^+) \leq \lambda\sigma_B(1-\sigma_F)\frac{\bar{w}-(1+\beta)u(x_0^-)}{\bar{w}-(1+\beta)u(x'^+)}\pi_B^H(x'^+)$$

$$\leq \lambda\sigma_B(1-\sigma_F)\frac{\bar{w}-(1+\beta)u(x_0^-)}{\bar{w}-(1+\beta)u(x'^+)}(\bar{w}-u(x'^+)) < \lambda\sigma_B(1-\sigma_F)(\bar{w}-u(x_0^-)),$$

where the last inequality follows from $\frac{\bar{w}-u(x_0^-)}{\bar{w}-u(x'^+)} > \frac{\bar{w}-(1+\beta)u(x_0^-)}{\bar{w}-(1+\beta)u(x'^+)}$, for $u(x'^+) < u(x_0^-)$, and $\beta > 0$. Also, any x'^+ such that $u(x'^+) > u(x_2^+)$ to borrowers with a positive signal at the bank will be accepted by all those borrowers but will result in a profit lower than $\lambda\sigma_B(1-\sigma_F)(\bar{w}-u(x_0^-))$ since $\pi_B^H(x'^+) < \pi_\zeta^H(x_2^+) = (1-\sigma_F)\pi_\zeta^H(H(x_0^-))$. Therefore, the bank has no profitable deviation from its offers to the borrowers with a positive signal.

b) **No Deviation for the Fintech:** First note that the bank's offers eliminate the possibility that the fintech earns positive profit from the borrowers with a negative signal at the fintech. For any $x^+ \in M_F^+$, the fintech's profit writes:

$$\Pi_F^+(x^+, \hat{X}_B) = \lambda\sigma_F(1-\sigma_B+\sigma_B F_B^+(x^+))\pi_F^H(x^+)$$

$$= \lambda\sigma_F\left(1-\sigma_B+\sigma_B(\frac{(1-\sigma_F)(\bar{w}-u(x_0^-))}{\sigma_B(\bar{w}-u(x^+))}-\frac{1-\sigma_B}{\sigma_B})\right)(\bar{w}-u(x^+))$$
$$= \lambda\sigma_F(1-\sigma_F)(\bar{w}-u(x_0^-)) = \lambda\sigma_F\pi_F^H(x_2^+).$$

Hence, the fintech is indifferent between all contracts in M_F^+. Any $x^{'+}$ such that $u(x^{'+}) > u(x_2^+)$ to borrowers with a positive signal at the fintech will be accepted by all those borrowers but will result in a profit lower than $\lambda\sigma_F\pi_F^H(x_2^+)$. Also, any $x^{'+}$ such that $u(x^{'+}) \leq u(x^-0)$ results in zero profit. Therefore, the fintech has no profitable deviation.

2. We show that, for $\sigma_F \geq \sigma_B$, the following strategies construct an equilibrium:

$$M_B^- = \{x^- \mid u(x_0^-) \leq u(x) \leq u(x_1^-)\},$$

$$M_B^+ = \{x^+ \mid u(x_1^-) \leq u(x) \leq u(x_1^+)\},$$

$$F_B^-(x^-) = \frac{\bar{w}-u(x_1^-)}{\bar{w}-u(x^-)}, \ \ \forall(x^+,x^-) \in M_B,$$

$$F_B^+(x^+) = 1-\frac{u(x_1^+)-u(x^+)}{\sigma_B(\bar{w}-u(x^+))}, \ \ \forall(x^+,x^-) \in M_B, \ \ \sigma_B \neq 0,$$

$$M_F^+ = \{x^+ \mid u(x_0^-) \leq u(x) \leq u(x_1^+)\},$$

$$M_F^- = \{x^- \mid u(\underline{x}^-)) \leq u(x) < u(x_0^-)\}, \ \ \text{for some } \underline{x}^- \text{ s.t } u(\underline{x}^-) < u(x_0^-),$$

$$F_F^+(x^+) = \begin{cases} \frac{(1-\sigma_F)(1+\beta)(u(x^+)-u(x_0^-))}{\sigma_F(\bar{w}-(1+\beta)u(x^+))}, & , \ u(x^+) \leq u(x_1^-) \\ \frac{1-\sigma_F}{\sigma_F}\left(\frac{(1+\beta)(u(x_1^-)-u(x_0^-))}{\bar{w}-(1+\beta)u(x_1^-)}+\frac{u(x^+)-u(x_1^-)}{\bar{w}-u(x^+)}\right) & , \ u(x^+) \geq u(x_1^-) \end{cases}$$
$$\forall(x^+,x^-) \in M_F, \ \ \sigma_F \neq 0,$$

and any proper CDF $F_F^-(x^-)$ such that:

$$F_F^-(x^-) \leq \frac{\bar{w}-(1+\beta)u(x_0^-)}{\bar{w}-(1+\beta)u(x^-)}, \ \ \forall(x^+,x^-) \in M_F.$$

a) **No Deviation for the Bank:** First consider bank's offers to borrowers with a negative signal. For any $x^- \in M_B^-$:

$$\Pi_B^-(x^-,\hat{X}_F) = \lambda(1-\sigma_B)(1-\sigma_F+\sigma_F F_F^+(x^-))\pi_B^H(x^-)$$

$$= \lambda(1-\sigma_B)(1-\sigma_F)(\bar{w}-(1+\beta)u(x_0^-)) = \lambda(1-\sigma_B)(1-\sigma_F)\pi_B^H(x_0^-).$$

Hence, the bank is indifferent between all contracts in M_B^-. Note that since $F_F^+(u(x^+))$ is larger for any x^- such that $u(x^-) \geq u(x_1^-)$, the bank's profit is strictly lower from offering any such contract. Also, $F_F^-(x^-)$ is the same as the previous case and eliminates the possibility of bank's deviation by offering contracts such that $u(x^-) < u(x_0^-)$. Therefore, the bank has no deviation from its offers to the borrowers with a negative signal. Now consider bank's offer to the borrowers with a positive signal. For any $x^+ \in M_B^+$:

$$\Pi_B^+(x^+, \hat{X}_F) = \lambda\sigma_B(1-\sigma_F+\sigma_F F_F^+(x^+))\pi_B^H(x^+) = \lambda\sigma_B(1-\sigma_F+\sigma_F F_F^+(x^+))(\bar{w}-u(x^+))$$
$$= \lambda\sigma_B\bigg(1-\sigma_F+\sigma_F F_F^+(u(x_1^-))\bigg)(\bar{w}-u(x_1^-)) = \lambda\sigma_B(\bar{w}-u(x_1^+)).$$

Hence, the bank is indifferent between all contracts in M_B^+. Note that since $F_F^+(u(x^+))$ is smaller for any x^+ such that $u(x^+) < u(x_1^-)$, the bank's profit is strictly lower from offering any such contract. Moreover, any contract x^+ such that $u(x^+) > u(x_1^+)$ results in a profit lower than $\lambda\sigma_B(\bar{w}-u(x_1^+))$. Also, as in the previous case, $F_F^-(x^-)$ eliminates the possibility of bank's deviation by offering any contract x^+ such that $u(x^+) < u(x_0^-)$.

b) **No Deviation for the Fintech:** Profit of the fintech from any offer x^+ such that $u(x_0^-) \leq u(x^+) \leq u(x_1^-)$ writes:

$$\Pi_F^+(x^+, \hat{X}_B) = \lambda\sigma_F(1-\sigma_B)F_B^-(x^+)(\bar{w}-u(x^+))$$
$$= \lambda\sigma_F(\bar{w}-u(x_1^+)) = \lambda\sigma_F(1-\sigma_B)(\bar{w}-u(x_1^-)).$$

Profit of the fintech from any offer x^+ such that $U(x_1^-) \leq u(x^+) \leq u(x_1^+)$ writes:

$$\Pi_F^+(x^+, \hat{X}_B) = \lambda\sigma_F(1-\sigma_B+\sigma_B F_B^+(x^+))(\bar{w}-u(x^+))$$
$$= \lambda\sigma_F(\bar{w}-u(x_1^+)) = \lambda\sigma_F(1-\sigma_B)(\bar{w}-u(x_1^-)).$$

Hence, the fintech is indifferent between all contracts in M_F. For any x such that $u(x) < u(x_0^-)$, fintech earns zero profit. For any x^+ such that $u(x^+) > u(x_1^+)$, fintech earns a lower per-borrower profit than $(\bar{w}-u(x_1^+))$ and does not attract more borrowers. Thus, fintech has no profitable deviation.

□

Proof. of Proposition 7:

We denote $u_1 \equiv u(x_1^-)$, and define $\delta \equiv \frac{\sigma_F-\sigma_B}{1-\sigma_B}$.
From the definition of x_1^- and x_1^+:

$$\bar{w}-(1+\beta)u_1 = \frac{1-\sigma_F}{1-\sigma_B}(\bar{w}-(1+\beta)u_0) \quad\rightarrow\quad u_1 = u_0 + (\frac{\bar{w}}{1+\beta}-u_0)\delta = u_0+\alpha\delta,$$

$$\bar{w}-u(x_1^+) = (1-\sigma_B)(\bar{w}-u_1) \quad\rightarrow\quad u(x_1^+) = u_0+\alpha\sigma_F+\frac{\beta}{1+\beta}\bar{w}\sigma_B.$$

a) **Profits:** From proposition 6:

$$\Pi_B(\sigma_B,\sigma_F) = \Pi_B^+(\sigma_B,\sigma_F) + \Pi_B^-(\sigma_B,\sigma_F)$$

$$= \begin{cases} \lambda(1-\sigma_F)(\bar{w}-u_0-(1-\sigma_B)\beta u_0) & ,\sigma_F<\sigma_B \\ \lambda(1-\sigma_B)\Big(\sigma_B(\bar{w}-u_0-\alpha\delta)+(1-\sigma_F)(\bar{w}-(1+\beta)u_0)\Big) & ,\sigma_F\geq\sigma_B \end{cases}$$

$$\Pi_F(\sigma_B,\sigma_F) = \Pi_F^+(\sigma_B,\sigma_F) + \Pi_F^-(\sigma_B,\sigma_F)$$

$$= \begin{cases} \lambda\sigma_F(1-\sigma_F)(\bar{w}-u_0) & ,\sigma_F<\sigma_B \\ \lambda\sigma_F(\bar{w}-u_0-\alpha\sigma_F-\frac{\beta}{1+\beta}\bar{w}\sigma_B) & ,\sigma_F\geq\sigma_B \end{cases}$$

b) **Welfare:** For $\sigma_F<\sigma_B$:

$$SI(\sigma_B,\sigma_F) = \lambda(1-\sigma_B)(1-\sigma_F)\beta u_0,\ \ \sigma_B<\sigma_F$$

$$\rightarrow\ \ W(\sigma_B,\sigma_F) = \lambda\Big(\bar{w}-(1-\sigma_B)(1-\sigma_F)\beta u_0\Big),\ \ \sigma_B<\sigma_F$$

For $\sigma_F\geq\sigma_B$, similar to the proof of proposition 2, we have:

$$SI(\hat{E}) = \int_{x_0^-}^{x_1^-} \lambda(1-\sigma_B)[1-\sigma+\sigma F_F^+(x^-)]\beta u(x^-).dF_B^-(x^-),$$

$$\mu_B^-(x_0^-) = \frac{\bar{w}-u(x_1^-)}{\bar{w}-u(x_0^-)},\ \text{ and }\ f_B^-(x) = \frac{\bar{w}-u(x_1^-)}{(\bar{w}-u(x))^2},\ \ x\neq x_0^-,$$

$$F_F^+(x) = \frac{(1-\sigma_F)(1+\beta)(u(x)-u(x_0^-))}{\sigma_F(\bar{w}-(1+\beta)u(x))}\ ,\ u(x)\in[u(x_0^-),u(x_1^-)]$$

Changing the variable to $u=u(x)$ and calculating the integral yields:

$$SI(\sigma_B,\sigma_F) = \lambda(1-\sigma_F)(1-\sigma_B)\beta u_0\frac{\bar{w}-u_1}{\bar{w}-u_0}$$

$$+\lambda(1-\sigma_F)(1-\sigma_B)(\bar{w}-u_1)(\bar{w}-(1+\beta)u_0)\frac{1}{\beta\bar{w}}\Bigg[\ln\left(\frac{(1-\sigma_B)(\bar{w}-u_1)}{(1-\sigma)(\bar{w}-u_0)}\right)+\frac{1}{\bar{w}-u_0}-\frac{1}{\bar{w}-u_1}\Bigg].$$

Further simplifying and replacing $A\equiv\bar{w}-u_0$, $\bar{w}-(1+\beta)u_0=\alpha(1+\beta)$, and $\bar{w}-u_1=A-\alpha\sigma$, we can write:

$$W(\sigma_B,\sigma_F) = \lambda\bar{w}-\lambda\Bigg[(1-\sigma_B)^2(1-\delta)\Bigg(\beta u_0-\alpha\delta+\frac{\alpha(1+\beta)}{\beta\bar{w}}(A-\alpha\delta)\ln(\frac{A-\alpha\delta}{A(1-\delta)})\Bigg)\Bigg]$$

$$,\sigma_B\leq\sigma_F.$$

1. For $\sigma_F < \sigma_B$, the borrowers' utility writes:

$$U(\sigma_B, \sigma_F) = W(\sigma_B, \sigma_F) - (\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) = \lambda[u_0 + (\bar{w} - u_0)\sigma_F^2].$$

which is strictly increasing in σ_F.

For $\sigma_F \geq \sigma_B$, we can write:

$$\frac{\partial}{\partial \sigma_F} U(\sigma_B, \sigma_F) = \frac{\partial}{\partial \sigma_F} W(\sigma_B, \sigma_F) - \frac{\partial}{\partial \sigma_F}(\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) =$$

$$\lambda\Bigg(2\alpha\sigma_B + (A + \alpha - 2\alpha\delta)\ln(\frac{A - \alpha\delta}{A(1-\delta)})\Bigg),$$

which is non-negative since $A + \alpha - 2\alpha\delta > 0$, and $\frac{A-\alpha\delta}{A(1-\delta)} \in [0, \infty)$.

2. From (a), it is straightforward that the bank's profit is continuous and strictly decreasing in $\sigma_F \in [0, 1]$.

Note that, for $\sigma_F < \sigma_B$, the fintech's profit can has a local maximum at $\sigma_F = \frac{1}{2}$, if $\sigma_B > \frac{1}{2}$. For $\sigma_F \geq \sigma_B$, we can write:

$$\frac{\partial \Pi_F(\sigma_B, \sigma_F)}{\partial \sigma_F} = \lambda\Bigg(A - \frac{\beta}{1+\beta}\bar{w}\sigma_B - 2\alpha\sigma_F\Bigg).$$

At $\sigma_F = 1$, we can have $\frac{\partial \Pi_F(\sigma_B, \sigma_F)}{\partial \sigma_F} < 0$, iff

$$\frac{1 - \beta(1 - \sigma_B)}{1 + 2\beta} > \frac{u_0}{\bar{w}}.$$

Note that $\frac{u_0}{\bar{w}} < 1$, while $\frac{1-\beta(1-\sigma_B)}{1+2\beta}$ approaches 1 for sufficiently small β (large γ). Therefore, there exists a threshold γ_F, such that for $\gamma > \gamma_F$, the fintech's profit has an inverse-U shape in σ_F.

3. For $\sigma_F < \sigma_B$:

$$\frac{\partial}{\partial \sigma_F}(\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) = \lambda\Bigg((1 - \sigma_B)\beta u_0 - 2(\bar{w} - u_0)\sigma_F\Bigg),$$

$$\rightarrow \frac{\partial}{\partial \sigma_F}(\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) < 0 \text{ iff } \frac{\beta u_0}{\bar{w} - u_0} < \frac{2\sigma_F}{1 - \sigma_B}.$$

Therefore, for any σ_B, the joint profit can have an inverse-U shape in $\sigma_F \in [0, \sigma_B]$, if β is sufficiently small, or equivalently if γ is larger than some threshold γ_{BF_1}.

For $\sigma_F \geq \sigma_B$:

$$\frac{\partial}{\partial \sigma_F}(\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) = \lambda\Bigg(\beta u_0(1 - \sigma_B) - 2\alpha\sigma_F\Bigg)$$

Hence, there exists a threshold γ_{BF_2}, such that for $\gamma > \gamma_{BF_2}$, β is sufficiently small such that the joint profit becomes inverse-U shaped in $\sigma_F \in [\sigma_B, 1]$.

4. It is straightforward that $W(\sigma_B, \sigma_F)$ is strictly increasing in σ_F for $\sigma_F < \sigma_B$. For $\sigma_F \geq \sigma_B$, we have:

$$W(\sigma_B, \sigma_F) = \lambda\bar{w} - \lambda\left[(1-\sigma_B)^2 \underbrace{(1-\delta)\left(\beta u_0 - \alpha\delta + \frac{\alpha(1+\beta)}{\beta\bar{w}}(A-\alpha\delta)\ln(\frac{A-\alpha\delta}{A(1-\delta)})\right)}_{\phi(\delta)}\right].$$

Note that the function $\phi(\delta)$ is exactly the same as $SI(\sigma)$ in the proof of proposition 2 (Note that $\delta = \sigma_F$ for $\sigma_B = 0$). Hence, the same result as in proposition 2 holds; there exist a threshold γ_W, such that for $\gamma > \gamma_W$, total welfare is bimodal in $\sigma_F \in [\sigma_B, 1]$.

□

Proof. of Proposition 8:

1. For $\sigma_B \in [0, \sigma_F]$:

$$\frac{\partial}{\partial\sigma_B}U(\sigma_B, \sigma_F) = \frac{\partial}{\partial\sigma_B}W(\sigma_B, \sigma_F) - \frac{\partial}{\partial\sigma_B}(\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) =$$

$$\lambda\left(2\sigma_B(2u_0 + \frac{\beta\bar{w}}{1+\beta}) + \frac{\alpha(1+\beta)}{\beta\bar{w}}(1-\sigma_F)(A-\alpha)\ln(\frac{A-\alpha\delta}{A(1-\delta)})\right) > 0.$$

Therefore $U(\sigma_B, \sigma_F)$ is increasing in $\sigma_B \in [0, \sigma_F]$.

From the proof of proposition 7, for $\sigma_B \in [\sigma_F, 1]$, the borrowers' utility writes $U(\sigma_B, \sigma_F) = \lambda[u_0 + (\bar{w} - u_0)\sigma_F^2]$, which is constant in σ_B.

2. From (a) in the proof of proposition 7, it is straightforward that the fintech's profit $\pi_F(\sigma_B, \sigma_F)$ is strictly decreasing in $\sigma_B \in [0, \sigma_F]$, and constant for

$$\sigma_B \in [\sigma_F, 1]$$

.

Considering the bank's profit, for $\sigma_B \in [0, \sigma_F]$:

$$\frac{\partial}{\partial\sigma_B}\Pi_B(\sigma_B, \sigma_F) = \lambda[\beta(u_0 + \alpha\sigma_F) - 2\sigma_B(2u_0 + \frac{\beta\bar{w}}{1+\beta})].$$

Note that for sufficiently low β, we can have $\frac{\partial}{\partial\sigma_B}\Pi_B(\sigma_B, \sigma_F) < 0$ for any $\sigma_F > 0$. Hence, there exists a threshold γ_B, such that for $\gamma > \gamma_B$ the bank's profit is inverse-U shaped in $\sigma_B \in [0, \sigma_F]$.

Also, for $\sigma_B \in [\sigma_F, 1]$, the bank's profit writes $\Pi_B(\sigma_B, \sigma_F) = \lambda(1-\sigma_F)(\bar{w} - u_0 - (1 - \sigma_B)\beta u_0)$, which is strictly increasing in σ_B.

3. For $\sigma_B \in [0, \sigma_F]$:

$$\frac{\partial}{\partial \sigma_B}(\Pi_B(\sigma_B, \sigma_F) + \pi_F(\sigma_B, \sigma_F)) = \lambda[\beta u_0(1-\sigma_F) - 2\sigma_B(2u_0 + \frac{\beta \bar{w}}{1+\beta}),]$$

$$\rightarrow \quad \frac{\partial}{\partial \sigma_B}(\Pi_B(.) + \pi_F(.)) < 0 \quad \text{iff} \quad \beta u_0(1-\sigma_F) < 2\sigma_B(2u_0 + \frac{\beta \bar{w}}{1+\beta}).$$

   The last inequality can hold for any $\sigma_B < \sigma_F$, if β is sufficiently small, i.e. γ is large enough. Hence, there exists a threshold γ^b_{BF}, such that for $\gamma > \gamma^b_{BF}$ the joint profit is inverse-U shaped in $\sigma_B \in [0, \sigma_F]$.

   For $\sigma_B \in [\sigma_F, 1]$, since the bank's profit is strictly increasing and the fintech's profit is constant in σ_B, the lenders' joint profit must be strictly increasing.

4. For $\sigma_B \in [0, \sigma_F]$:

$$\frac{\partial}{\partial \sigma_B} W(\sigma_B, \sigma_F) = \lambda\Bigg((1-\sigma_F)\beta u_0 + \frac{\alpha(1+\beta)}{\beta \bar{w}}(1-\sigma_F)(A-\alpha)\ln(\frac{A-\alpha\delta}{A(1-\delta)})\Bigg) > 0.$$

   For $\sigma_B \in [\sigma_F, 1]$: Since the joint profit is strictly increasing and the borrowers' utility is constant in σ_B, total welfare must be strictly increasing.

$\square$

Proof. of Proposition 9:

We define x^-_N such that:

$$\pi^H_B(x^-_N) = (1-\sigma)^N \pi^H_B(x^-_0).$$

Consider the following strategies:

1. The bank offers randomized contracts from the set $M_B = \{x^- \mid u(x^-_0) \leq u(x^-) \leq u(x^-_N)\}$ according to:

$$F_B(x^-) = (\frac{1}{1-\sigma})^{N-1}\left(\frac{\bar{w} - u(x^-_N)}{\bar{w} - u(x^-)}\right)\left(\frac{\bar{w} - (1+\beta)u(x^-)}{\bar{w} - (1+\beta)u(x^-_0)}\right)^{\frac{N-1}{N}}.$$

2. Each fintech offer random contracts to borrowers with positive signals from the set $M^+_F = \{x^+ \mid u(x^-_0) \leq u(x^-) \leq u(x^-_N)\}$ according to:

$$F_F(x^+) = (\frac{1-\sigma}{\sigma})\left[\left(\frac{\bar{w} - (1+\beta)u(x^-_0)}{\bar{w} - (1+\beta)u(x^-)}\right)^{\frac{1}{N}} - 1\right],$$

and offers random contracts to borrowers with negative signal from the set $M_F^- = \{x^+ \mid u(\underline{x}^-) \leq u(x^-) \leq u(x_0^-)\}$, for some $\underline{x}^-$ such that $u(\underline{x}^-) < u(x_0^-)$, according to any proper CDF $F_F^-(\cdot)$ such that:

$$F_F^-(x^-) \leq \left(\frac{\bar{w} - (1+\beta)u(x_0^-)}{\bar{w} - (1+\beta)u(x^-)} \right)^{\frac{1}{N}}.$$

We show that these strategies constitute an equilibrium.

No Deviation for the Bank. The bank's profit from any contract $x^- \in M_B$ writes:

$$\Pi_B(x^-) = \lambda(1 - \sigma + \sigma F_F^+(x^-))^N \pi_B^H(x^-) = \lambda(1 - \sigma + \sigma F_F^+(x^-))^N (\bar{w} - (1+\beta)u(x^-)).$$

Substituting

$$(1 - \sigma + \sigma F_F^+(x^-))^N = (1-\sigma)^N \left(\frac{\bar{w} - (1+\beta)u(x_0^-)}{\bar{w} - (1+\beta)u(x^-)} \right),$$

we obtain

$$\Pi_B(x^-) = \lambda(1-\sigma)^N (\bar{w} - (1+\beta)u(x_0^-)) = \lambda(1-\sigma)^N \pi_B^H(x_0^-).$$

The bank is therefore indifferent between all contracts in M_B. Any contract x^- with $u(x^-) > u(x_N^-)$ earns a profit strictly below $\lambda(1-\sigma)^N \pi_B^H(x_0^-)$. Moreover, if the bank offers any contract x^- with $u(x^-) < u(x_0^-)$, its profit writes:

$$\Pi_B(x^-) = \lambda \bigg((1-\sigma) F_F^-(x^-) \bigg)^N \pi_B^H(x^-) \leq \lambda(1-\sigma)^N (\bar{w} - (1+\beta)u(x_0^-)) = \lambda(1-\sigma)^N \pi_B^H(x_0^-).$$

Hence the bank has no profitable deviation.

No Deviation for the Fintech. Fintechs earn zero profit from their offers to borrowers with a negative signal and cannot profitably attract them. For any offer $x^+ \in M_F^+$, a fintech's profit writes:

$$\Pi_F^+(x^+) = \lambda \sigma F_B(x^+) \bigg(1 - \sigma + \sigma F_F^+(x^+) \bigg)^{N-1} \pi_F^H(x^+).$$

Substituting for $F_B(x^+)$ and $F_F^+(x^+)$, we obtain

$$\Pi_F^+(x^+) = \lambda\sigma(\bar{w} - u(x_N^-)) = \lambda\sigma\pi_F^H(G(x_N^-)).$$

The fintechs are therefore indifferent between all contracts in M_F^+. Any contract x^+ with $u(x^+) < u(x_0^-)$ earns the fintechs zero profit. Any contract x^+ with $u(x^+) > u(x_N^-)$ earns strictly less than $\lambda\sigma\pi_F^H(G(x_N^-))$. Hence the fintechs have no profitable deviation.

$\square$